

Exploratory Data Analysis

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
sns.set_theme(style="whitegrid")
import duckdb as db
```

```
In [2]: df=pd.read_parquet("../data/cleaned/dataset.parquet")
df['index'] = df.index
```

```
In [3]: df.dtypes
```

```
Out[3]: market_id                                float64
created_at                                datetime64[us]
actual_delivery_time                        datetime64[us]
store_id                                    object
store_primary_category                      object
order_protocol                             int8
total_items                                int64
subtotal                                    int64
num_distinct_items                         int64
min_item_price                             float64
max_item_price                             float64
total_onshift_partners                     float64
total_busy_partners                        float64
total_outstanding_orders                   float64
created_at_year                            int32
created_at_month                           int8
created_at_day                             int8
created_at_hour                             int8
created_at_minute                           int8
created_at_second                           int8
actual_delivery_time_year                   int32
actual_delivery_time_month                  int8
actual_delivery_time_day                    int8
actual_delivery_time_hour                   int8
actual_delivery_time_minute                 int8
actual_delivery_time_second                 int8
created_at_month_name                      object
actual_delivery_time_month_name             object
created_at_day_of_week                     int8
actual_delivery_time_day_of_week            int8
created_at_day_name                         object
actual_delivery_time_day_name               object
created_at_week_number                     int8
actual_delivery_time_week_number            int8
created_at_week_of_month                   int16
actual_delivery_time_week_of_month          int16
delivery_time_seconds                       int64
delivery_time_minutes                       int64
index                                       int64
dtype: object
```

```
In [4]: df.head()
```

	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	...	actual_delivery_time_c
0	1.0	2015-02-06 16:54:17	2015-02-06 17:57:16	df263d996281d984952c07998dc54358	american	1	4	3441	4	557.0	...	
1	2.0	2015-02-10 16:19:25	2015-02-10 17:26:29	f0ade77b43923b38237db569b016ba25	mexican	2	1	1900	1	1400.0	...	
2	3.0	2015-01-22 15:09:28	2015-01-22 15:39:09	f0ade77b43923b38237db569b016ba25	other	1	1	1900	1	1900.0	...	
3	3.0	2015-02-03 15:51:45	2015-02-03 16:43:00	f0ade77b43923b38237db569b016ba25	other	1	6	6900	5	600.0	...	
4	3.0	2015-02-14 21:10:36	2015-02-14 21:50:26	f0ade77b43923b38237db569b016ba25	other	1	3	3900	3	1100.0	...	

5 rows × 39 columns

```
In [5]: df.describe()
```

	market_id	created_at	actual_delivery_time	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	max_item_price	total_onshift_partners	...	actual_delivery_time_
count	177544.000000	177544	177544	177544.000000	177544.000000	177544.000000	177544.000000	177544.000000	177544.000000	177544.000000	...	177544.0
mean	2.767601	2015-02-04 14:02:31.056284	2015-02-04 14:50:05.157701	2.897192	3.205588	2701.713919	2.675618	692.093454	1164.986471	45.719624	...	29.0
min	1.000000	2015-01-21 09:52:03	2015-01-21 10:28:11	1.000000	1.000000	0.000000	1.000000	1.000000	52.000000	1.000000	...	0.0
25%	2.000000	2015-01-28 20:10:10.750000	2015-01-28 20:56:42.750000	1.000000	2.000000	1420.000000	2.000000	300.000000	799.000000	18.000000	...	14.0
50%	2.000000	2015-02-04 20:58:15.500000	2015-02-04 21:58:34.500000	3.000000	3.000000	2228.000000	2.000000	599.000000	1095.000000	38.000000	...	29.0
75%	4.000000	2015-02-11 19:19:06.250000	2015-02-11 19:59:20.250000	4.000000	4.000000	3421.000000	3.000000	943.000000	1397.000000	66.000000	...	44.0
max	6.000000	2015-02-18 00:30:44	2015-02-19 17:15:31	7.000000	411.000000	26800.000000	20.000000	14700.000000	14700.000000	171.000000	...	59.0
std	1.329823	NaN	NaN	1.511135	2.674640	1829.211488	1.624410	516.496735	559.219301	34.271843	...	17.0

8 rows × 33 columns

Univariate Analysis

```
In [78]: print(f"There are {df.shape[0]} rows in the dataset")
print(f"First orders were placed on {df['created_at'].min()} and last orders were placed on {df['created_at'].max()}")
# print number of days
print(f"There are {df['created_at'].dt.date.nunique()} unique days in the dataset")
```

There are 177544 rows in the dataset
First orders were placed on 2015-01-21 09:52:03 and last orders were placed on 2015-02-18 00:30:44
There are 29 unique days in the dataset

Market ID

```
In [26]: plt.figure(figsize=(10, 6))
df["market_id"].value_counts().plot(kind="bar", title="Number of orders per market");
plt.xlabel("Market ID")
plt.ylabel("Number of orders")
plt.show();
```

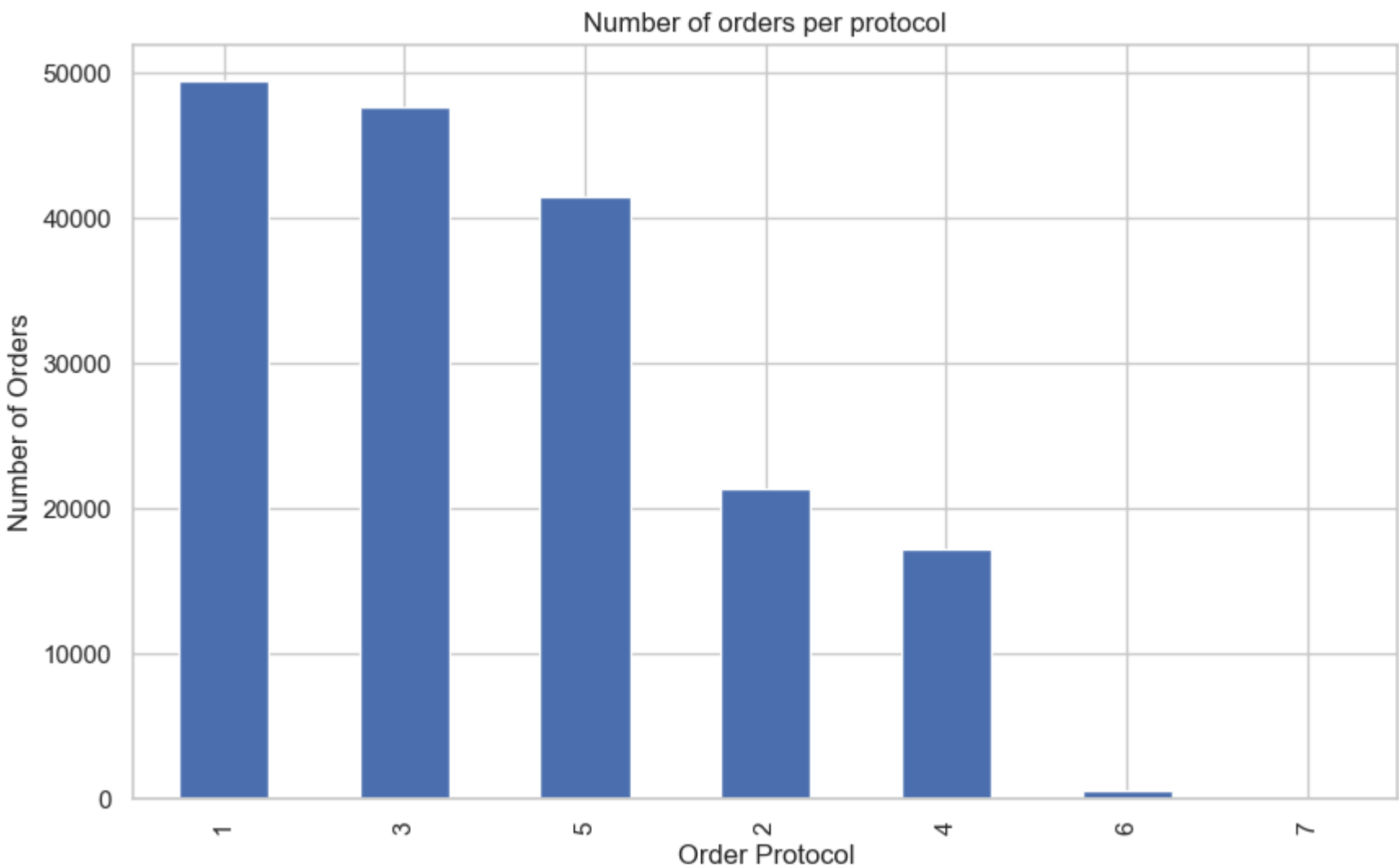


Observations

- Majority of the orders are from market id 2
- Market id 6 has the least number of orders

Order Protocol

```
In [27]: plt.figure(figsize=(10, 6))
df["order_protocol"].value_counts().plot(kind="bar", title="Number of orders per protocol");
plt.xlabel("Order Protocol")
plt.ylabel("Number of Orders");
```



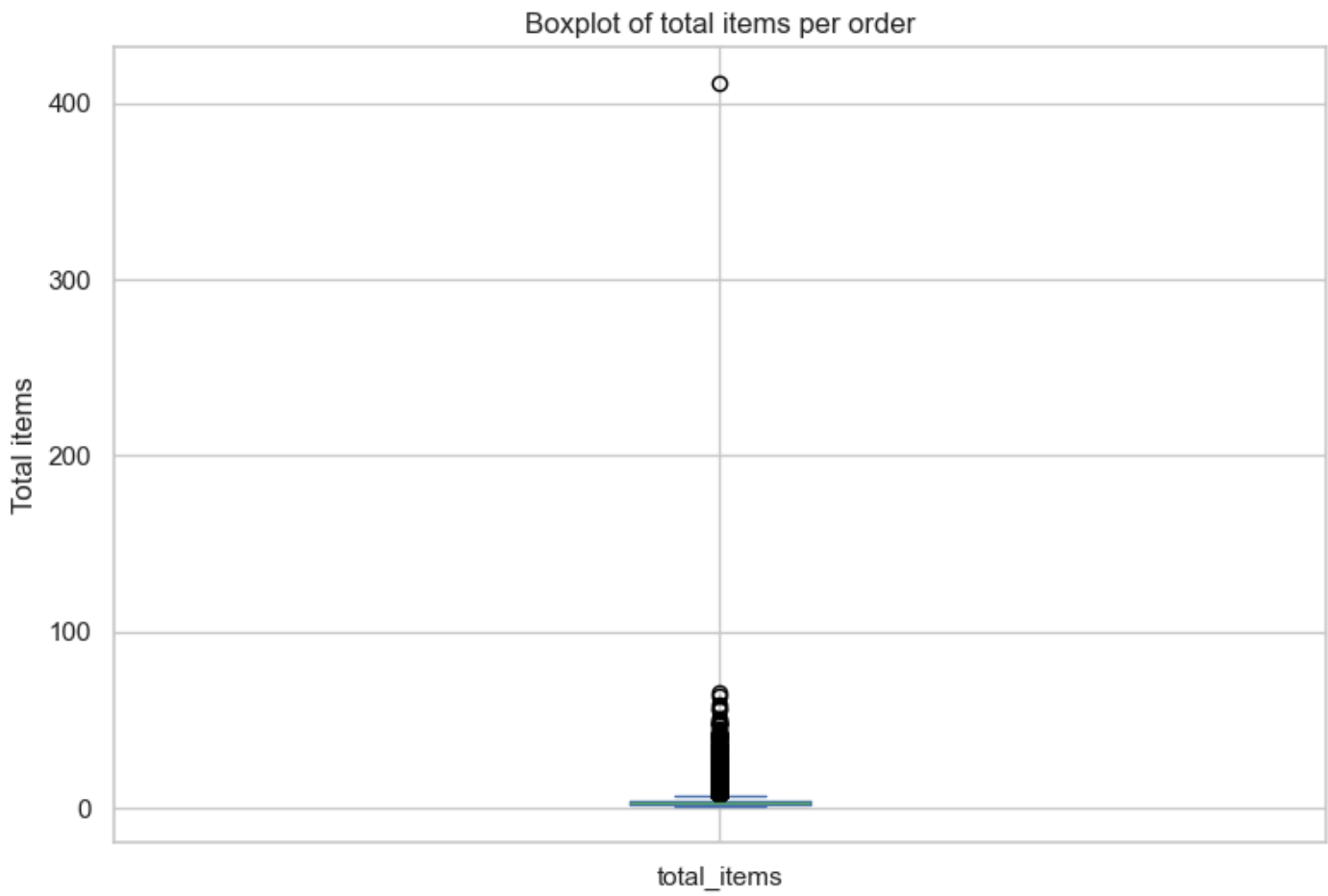
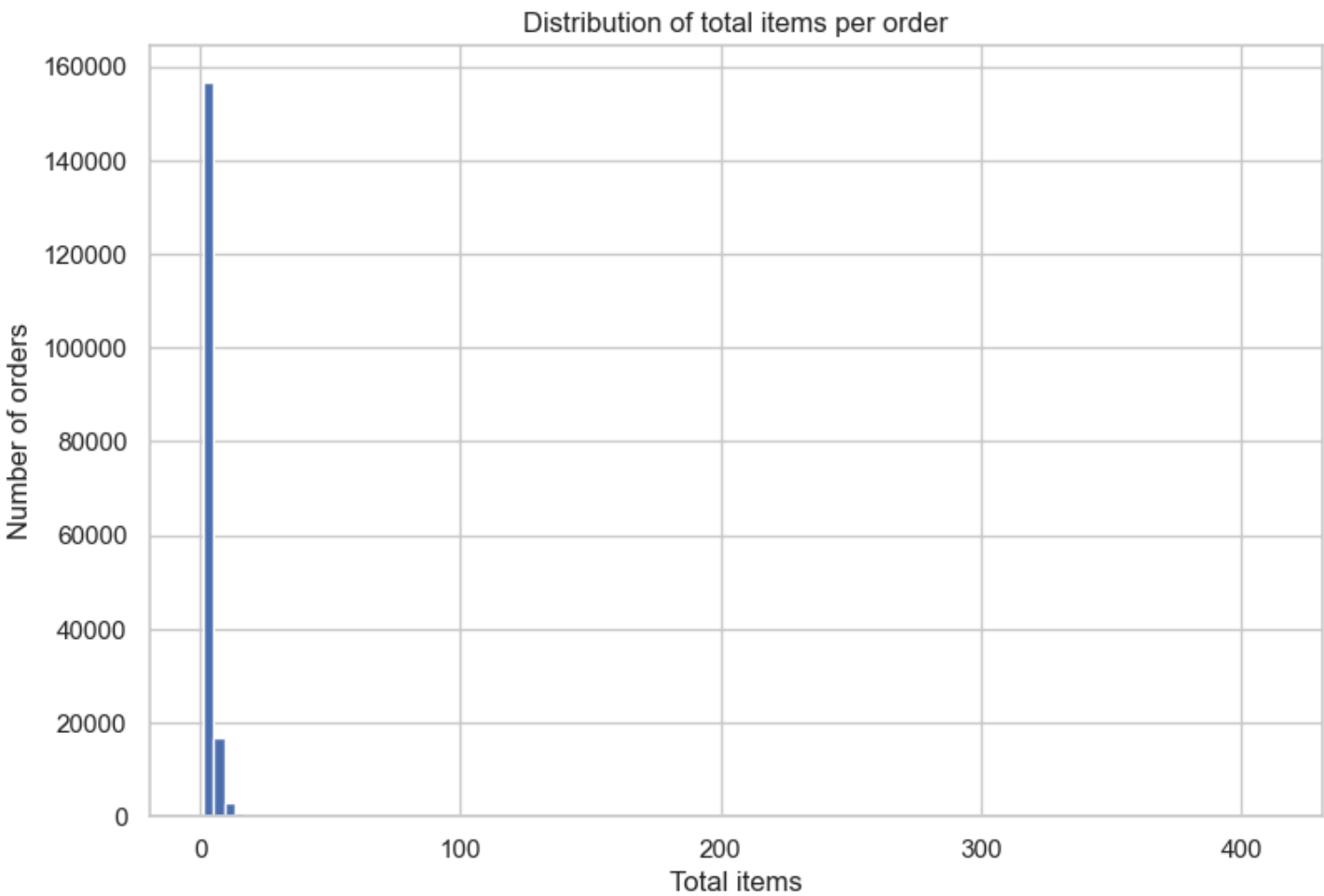
Observations

- Majority of the orders are from Order Protocol 1
- Order Protocol 7 has the least number of orders

Total Items

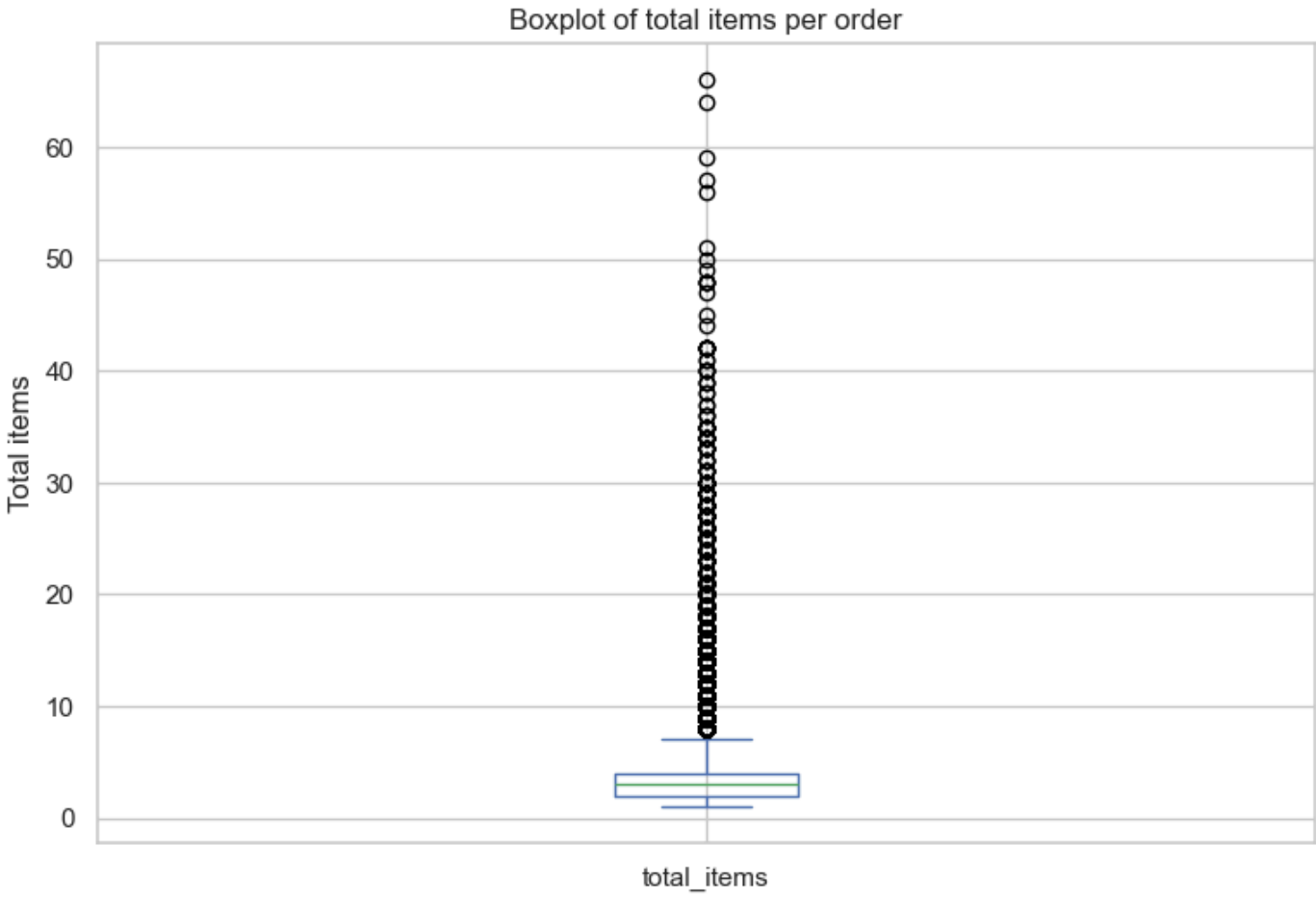
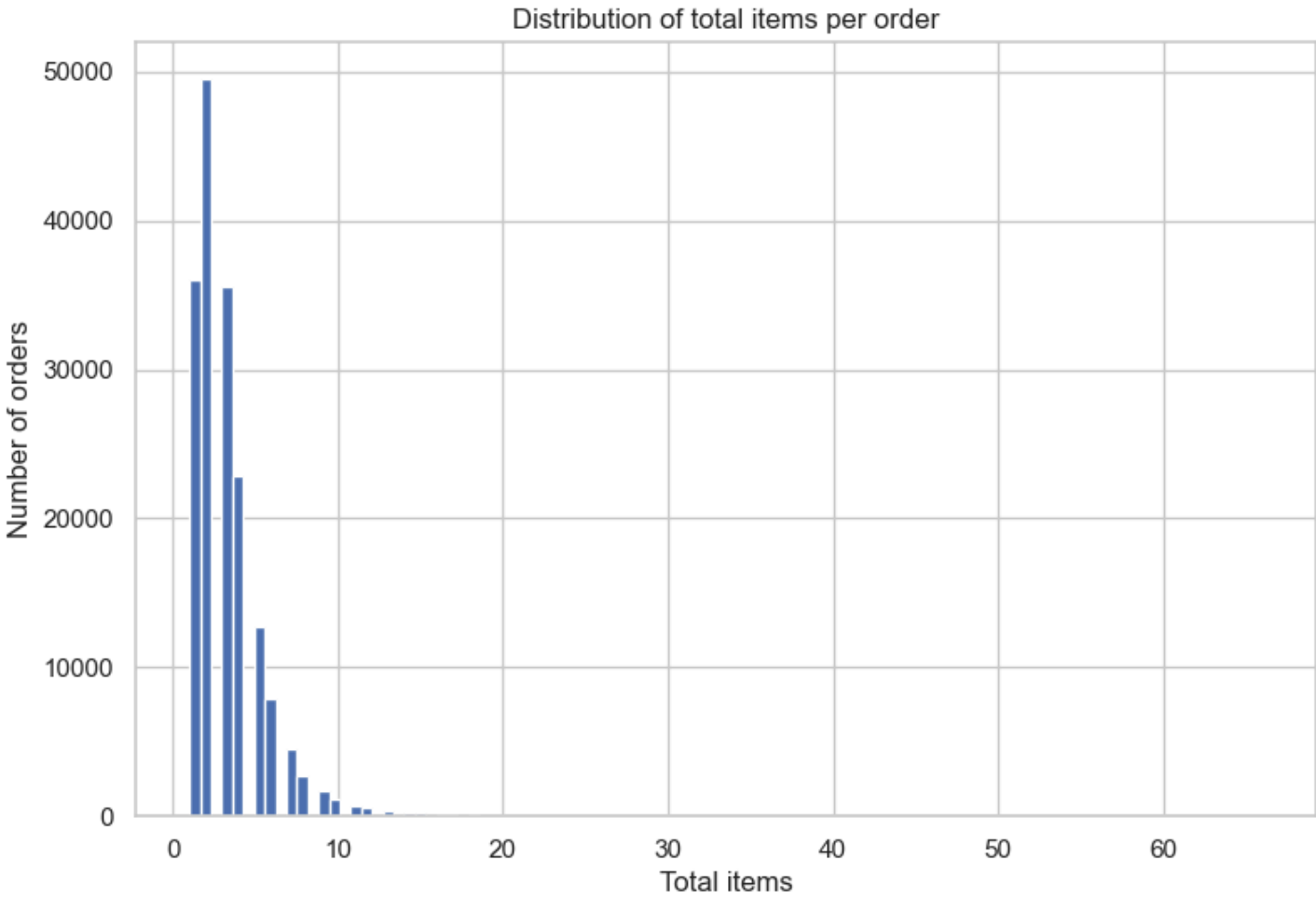
```
In [31]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

df["total_items"].hist(bins=100, ax=ax[0])
ax[0].set_title("Distribution of total items per order")
ax[0].set_xlabel("Total items")
ax[0].set_ylabel("Number of orders")
df["total_items"].plot(kind="box", ax=ax[1])
ax[1].set_title("Boxplot of total items per order")
ax[1].set_ylabel("Total items")
plt.show();
```



- We can see that there is an outlier in the data

```
In [8]: fig, ax = plt.subplots(1,2, figsize=(20, 6))
        _df[df["total_items"]<100]
        _["total_items"].hist(bins=100, ax=ax[0])
        ax[0].set_title("Distribution of total items per order")
        ax[0].set_xlabel("Total items")
        ax[0].set_ylabel("Number of orders")
        _["total_items"].plot(kind="box", ax=ax[1])
        ax[1].set_title("Boxplot of total items per order")
        ax[1].set_ylabel("Total items")
        plt.show();
```



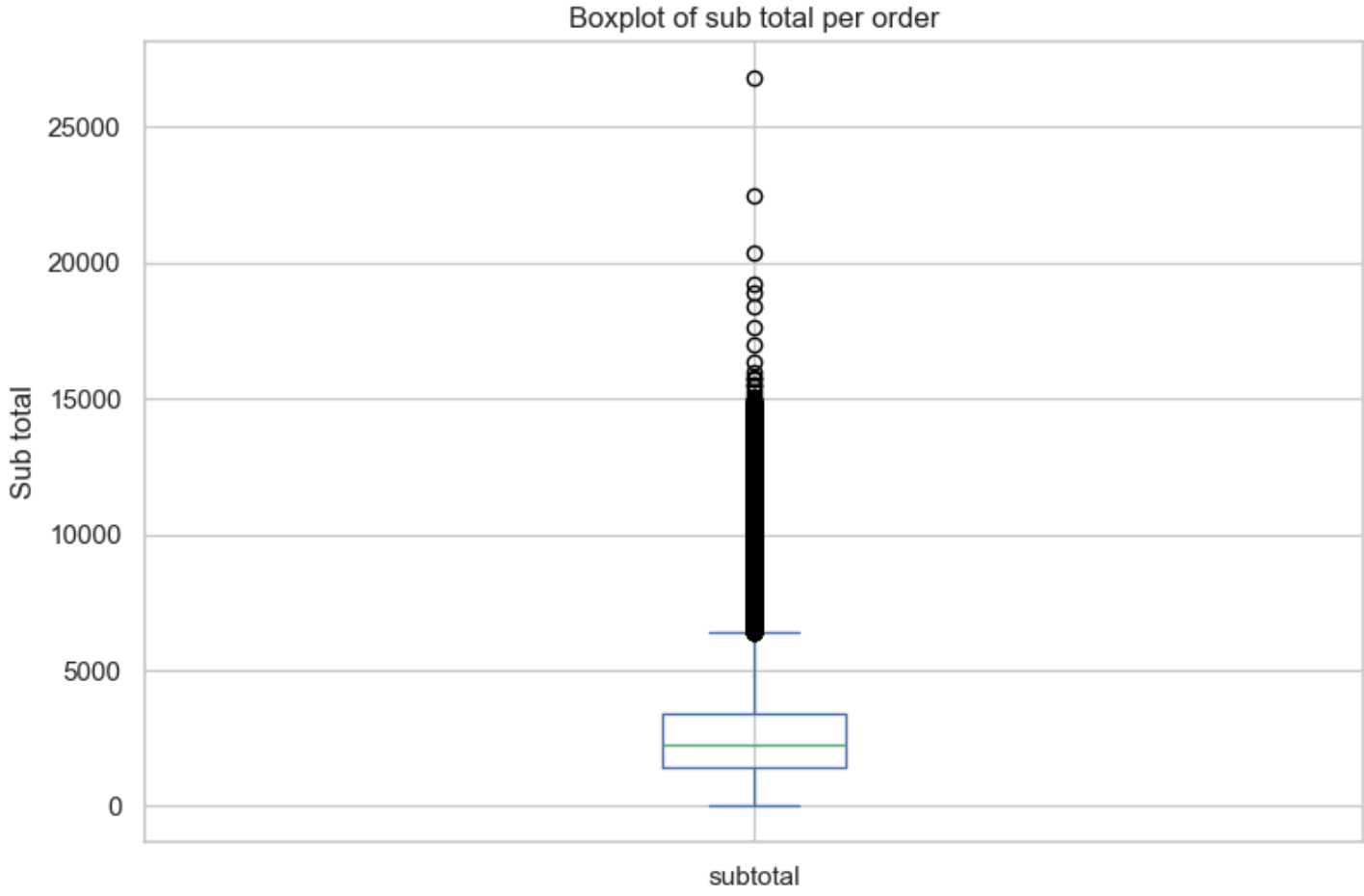
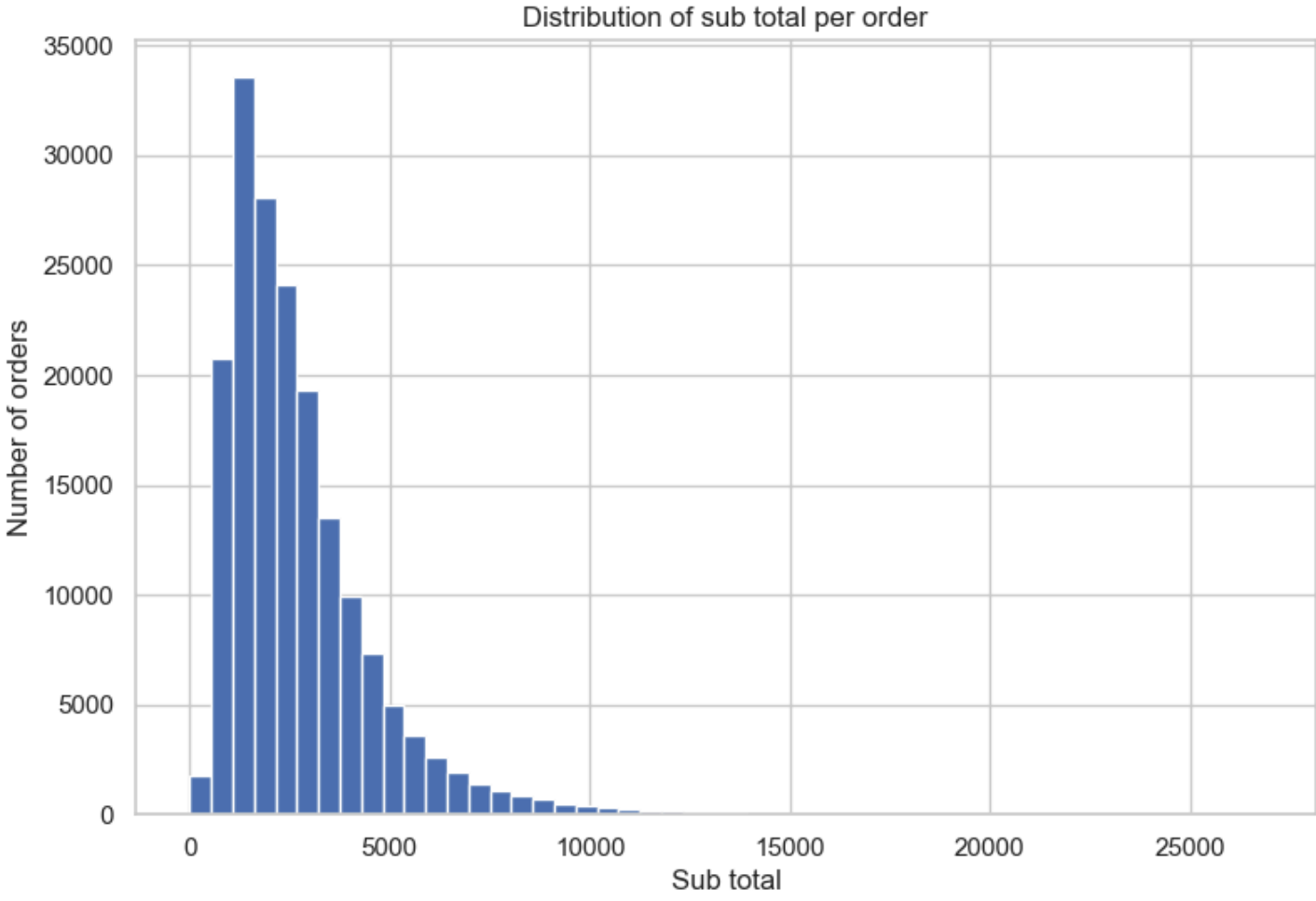
Observations

- From above plot we can see that most of the total items count is between 0 and 15

Sub Total

```
In [6]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

df["subtotal"].hist(bins=50, ax=ax[0])
ax[0].set_title("Distribution of sub total per order")
ax[0].set_xlabel("Sub total")
ax[0].set_ylabel("Number of orders")
df["subtotal"].plot(kind="box", ax=ax[1])
ax[1].set_title("Boxplot of sub total per order")
ax[1].set_ylabel("Sub total")
plt.show();
```



Observations

- We can see that most of the subtotal values are under 3000

Min Price and Max Price

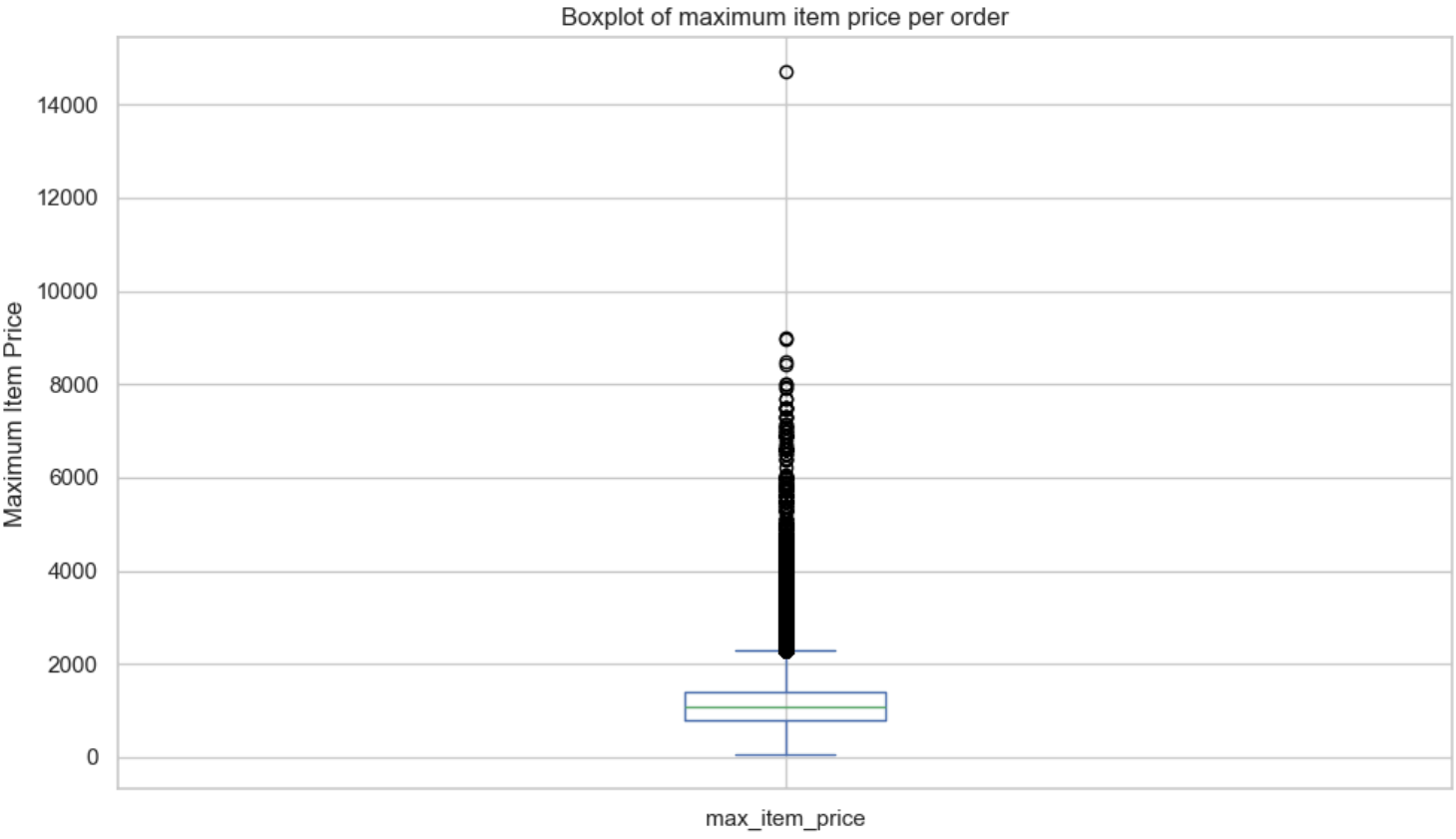
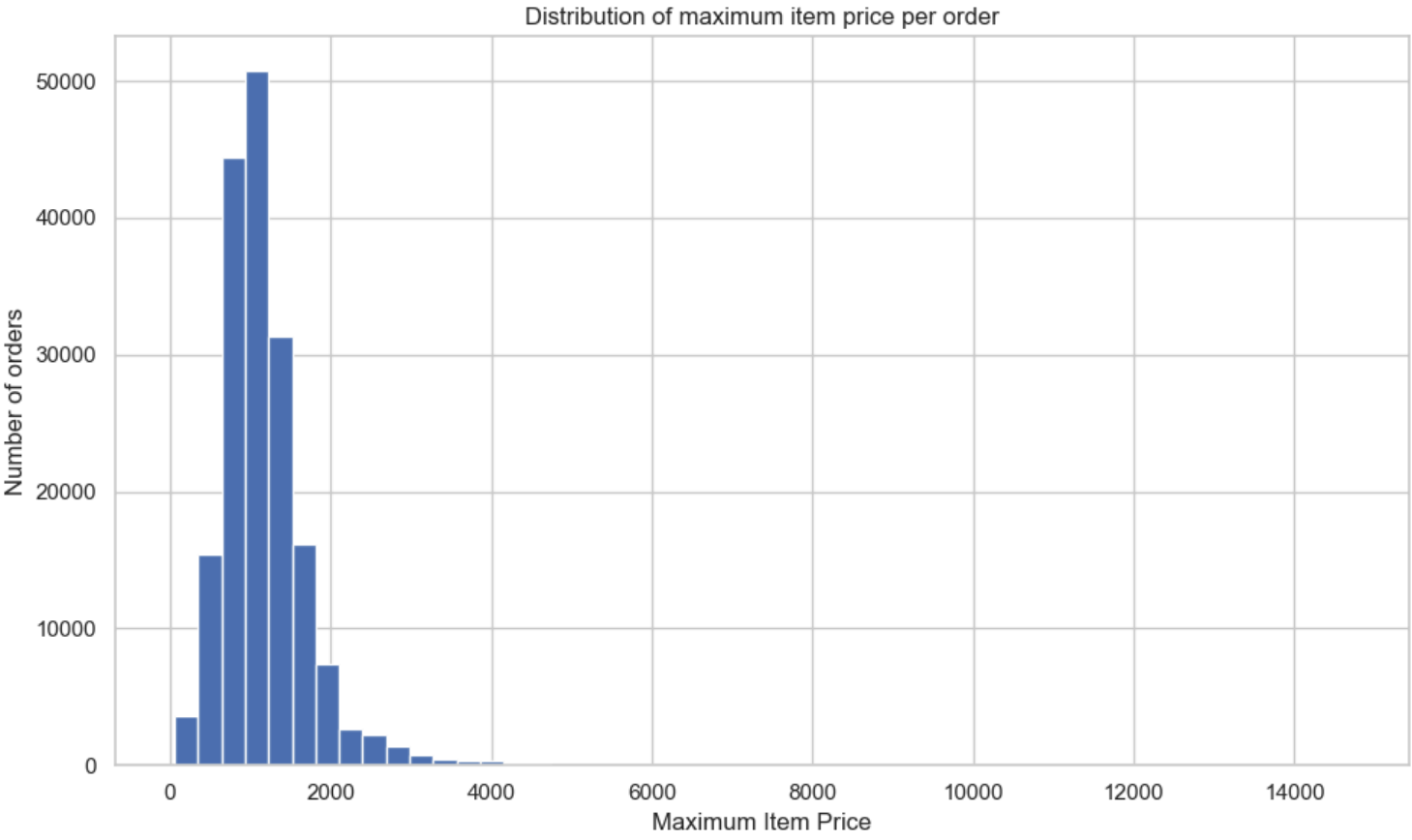
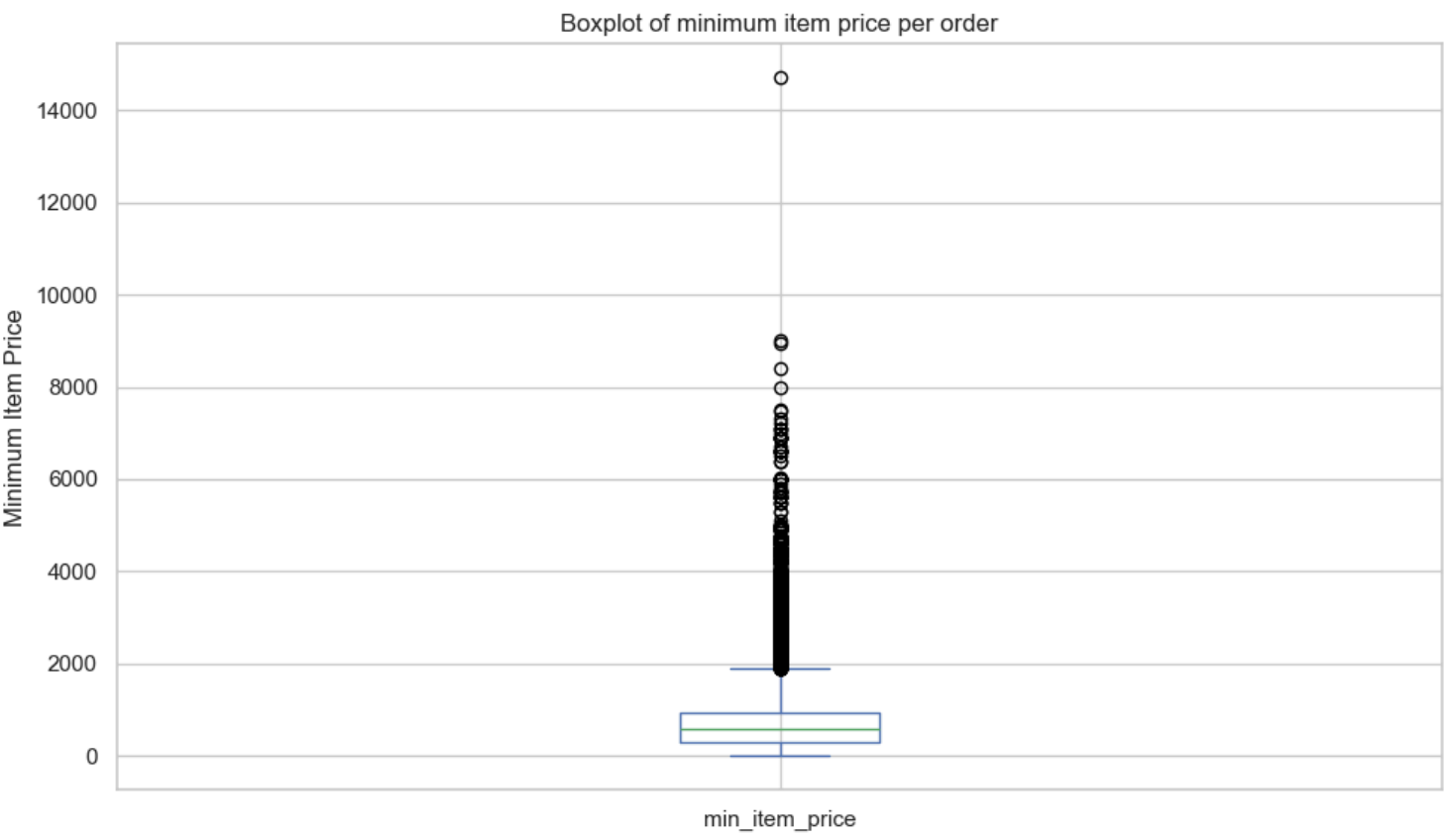
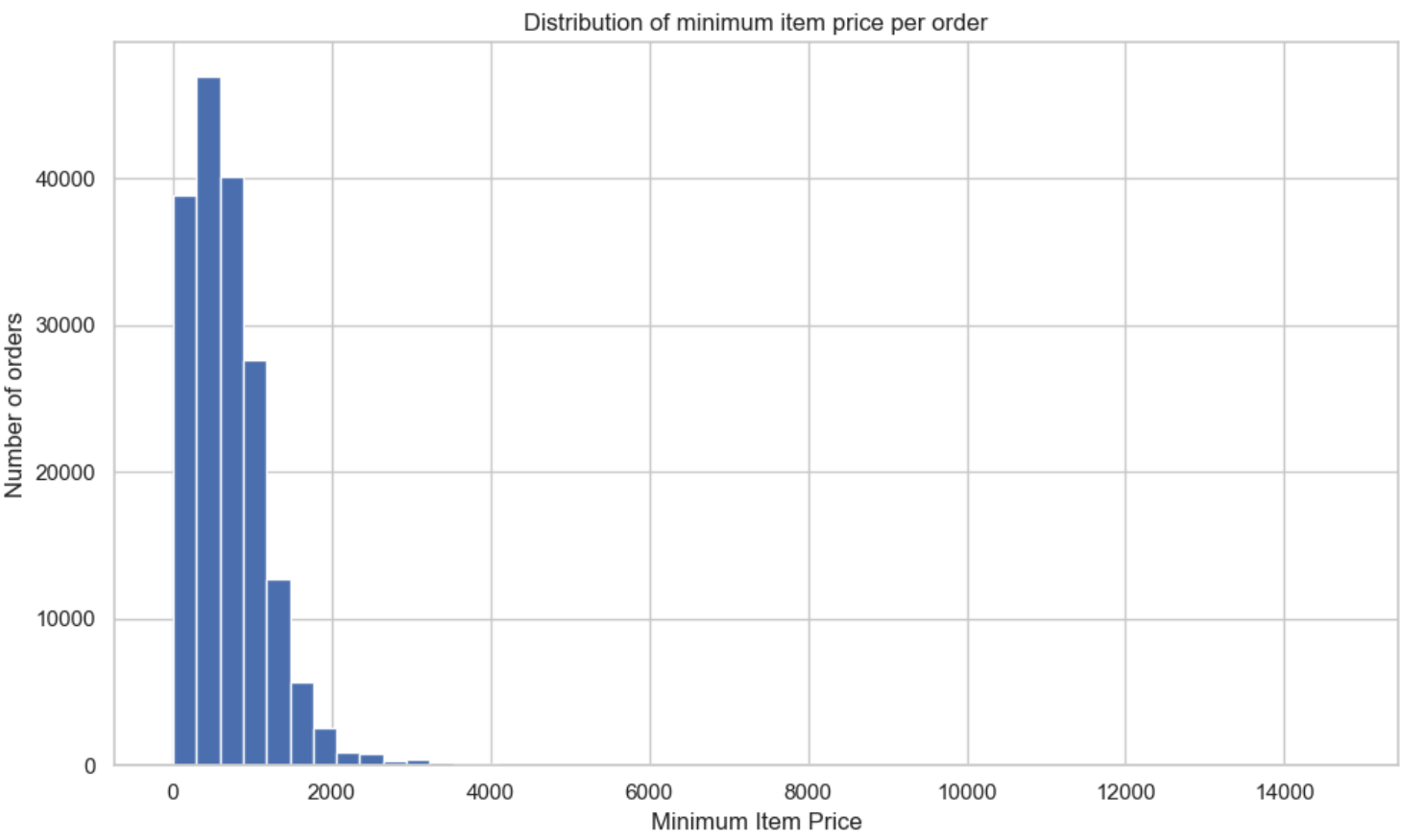
```
In [5]: fig, ax = plt.subplots(2,2, figsize=(20, 12))

df["min_item_price"].hist(bins=50, ax=ax[0][0])
ax[0][0].set_title("Distribution of minimum item price per order")
ax[0][0].set_xlabel("Minimum Item Price")
ax[0][0].set_ylabel("Number of orders")

df["min_item_price"].plot(kind="box", ax=ax[0][1])
ax[0][1].set_title("Boxplot of minimum item price per order")
ax[0][1].set_ylabel("Minimum Item Price")

# max_item_price
df["max_item_price"].hist(bins=50, ax=ax[1][0])
ax[1][0].set_title("Distribution of maximum item price per order")
ax[1][0].set_xlabel("Maximum Item Price")
ax[1][0].set_ylabel("Number of orders")
df["max_item_price"].plot(kind="box", ax=ax[1][1])
ax[1][1].set_title("Boxplot of maximum item price per order")
ax[1][1].set_ylabel("Maximum Item Price")

plt.tight_layout()
plt.show();
```

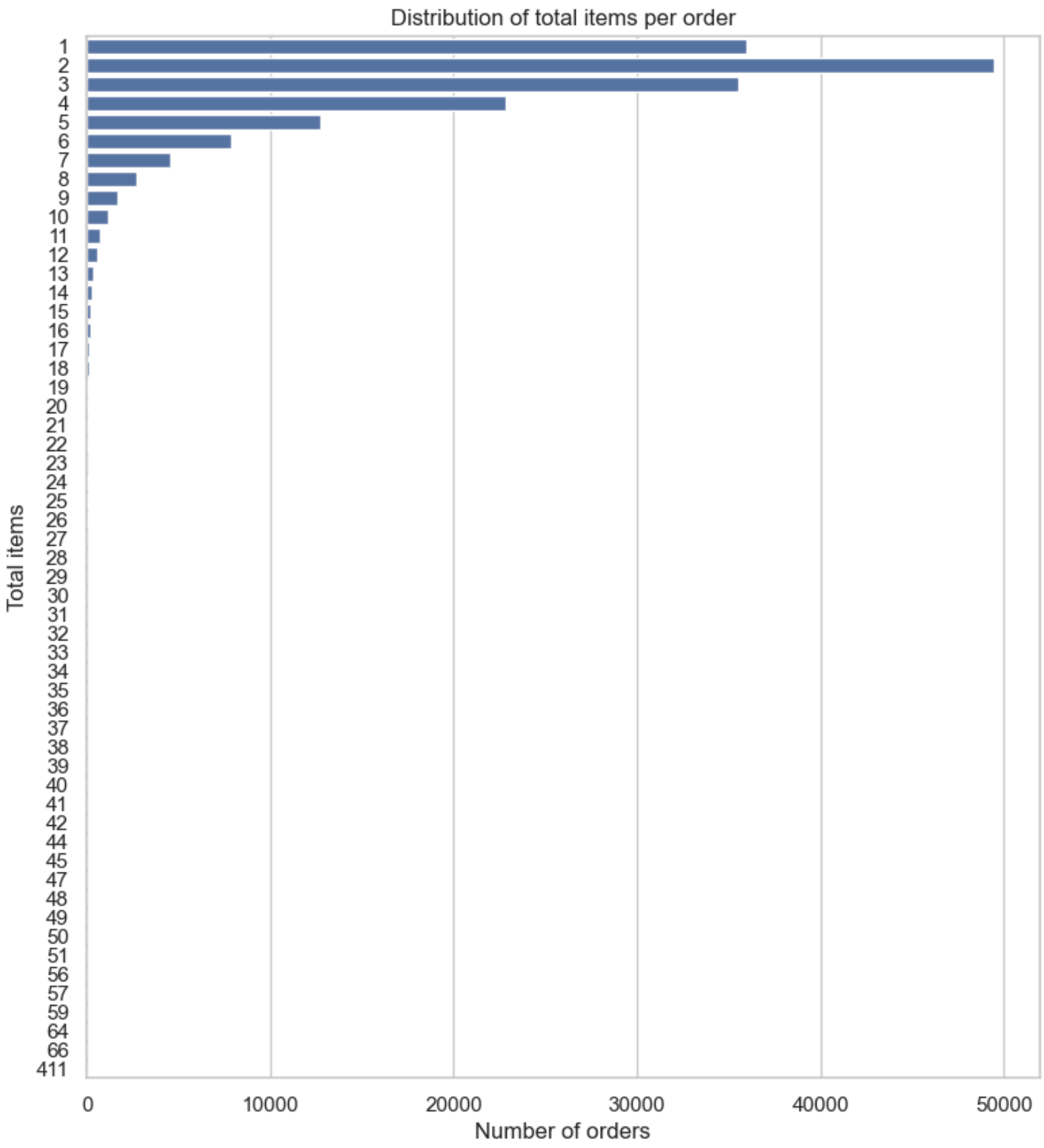
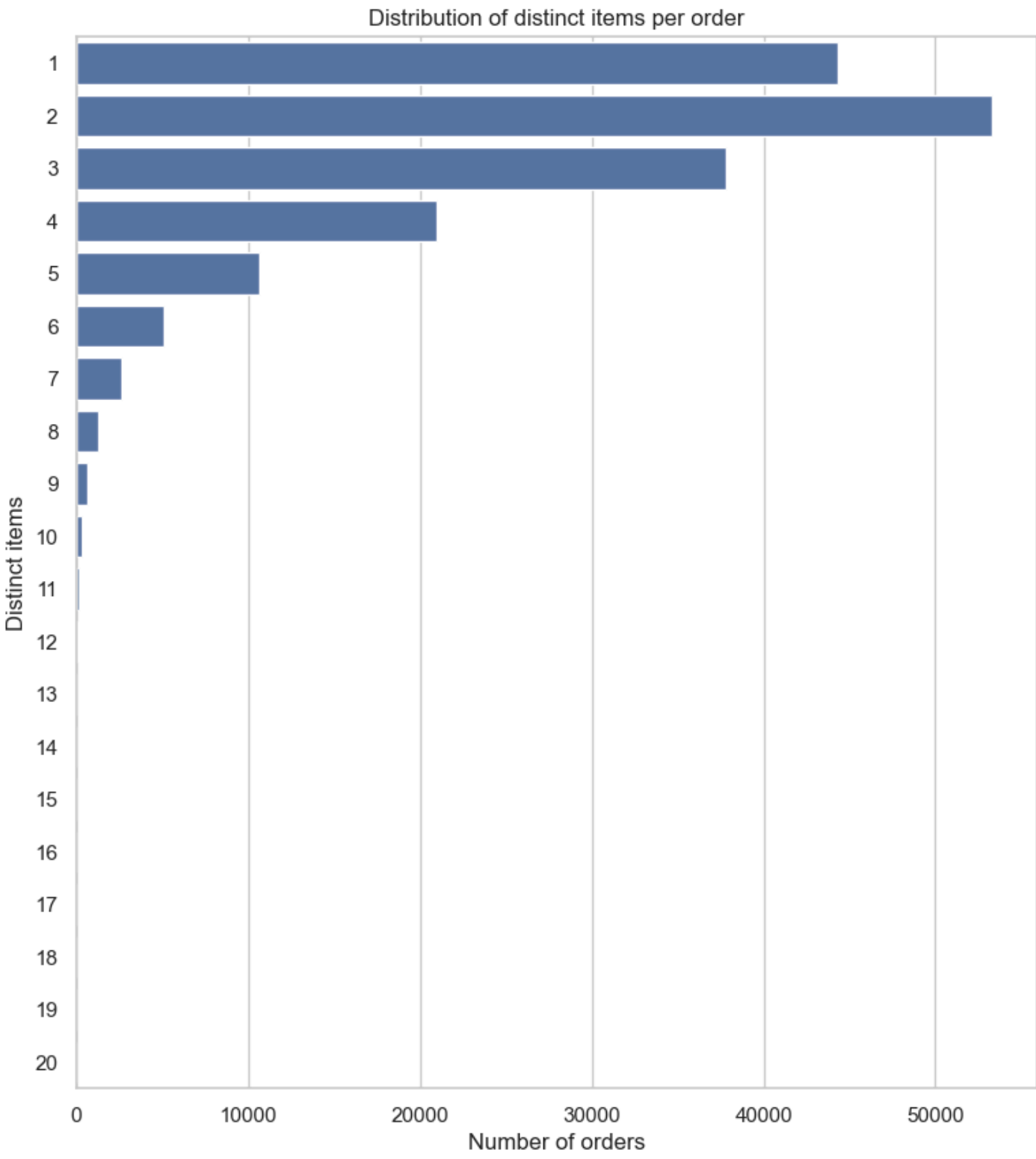



Observations

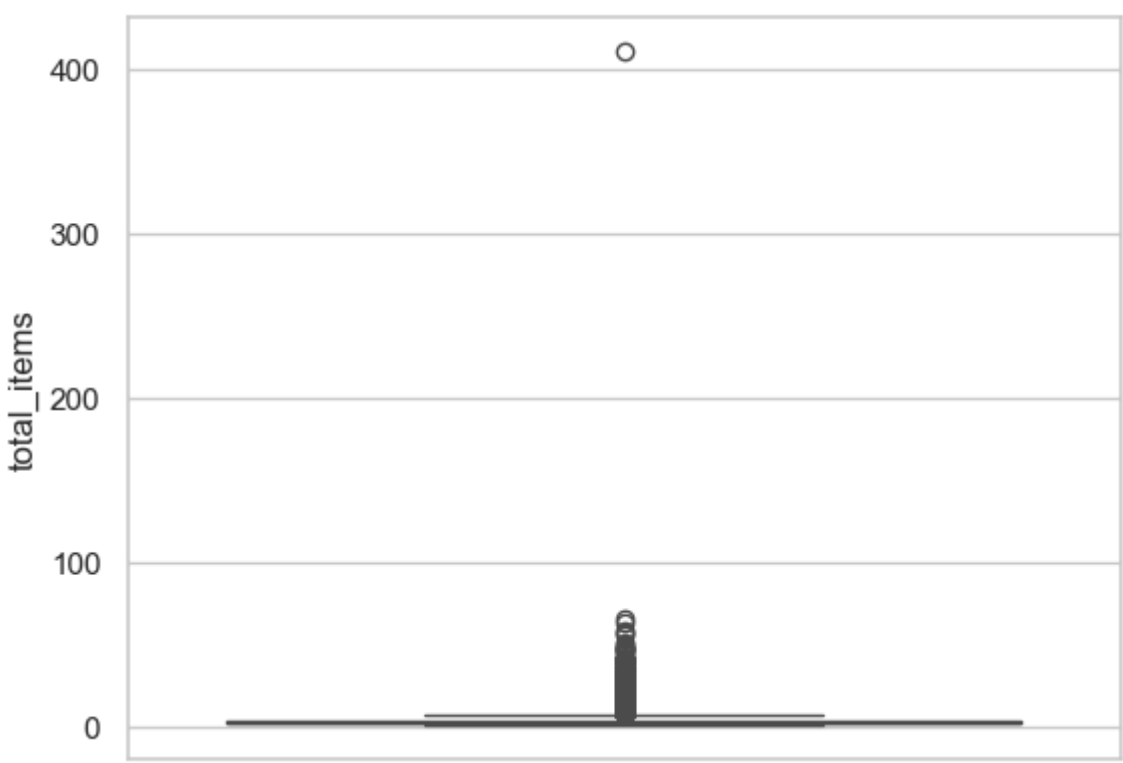
Distinct items

```
In [15]: fig, ax = plt.subplots(1, 2, figsize=(20, 10))
sns.countplot(data=df, y="num_distinct_items", ax=ax[0])
ax[0].set_title("Distribution of distinct items per order")
ax[0].set_ylabel("Distinct items")
ax[0].set_xlabel("Number of orders")

sns.countplot(data=df, y="total_items", ax=ax[1])
ax[1].set_title("Distribution of total items per order")
ax[1].set_ylabel("Total items")
ax[1].set_xlabel("Number of orders");
```



```
In [ ]:
```



Observations

Total OnShift Partners & Total Busy Partners

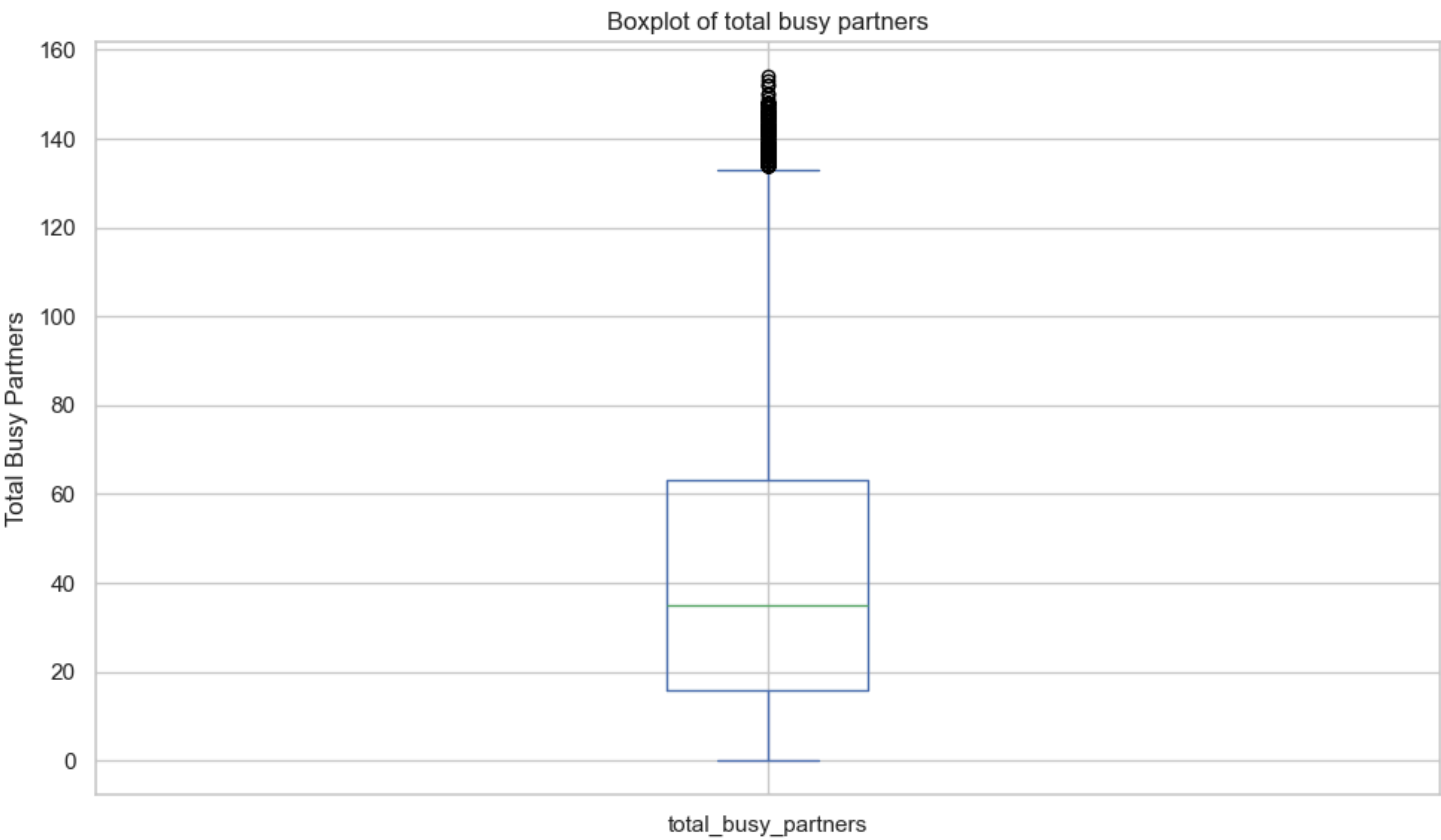
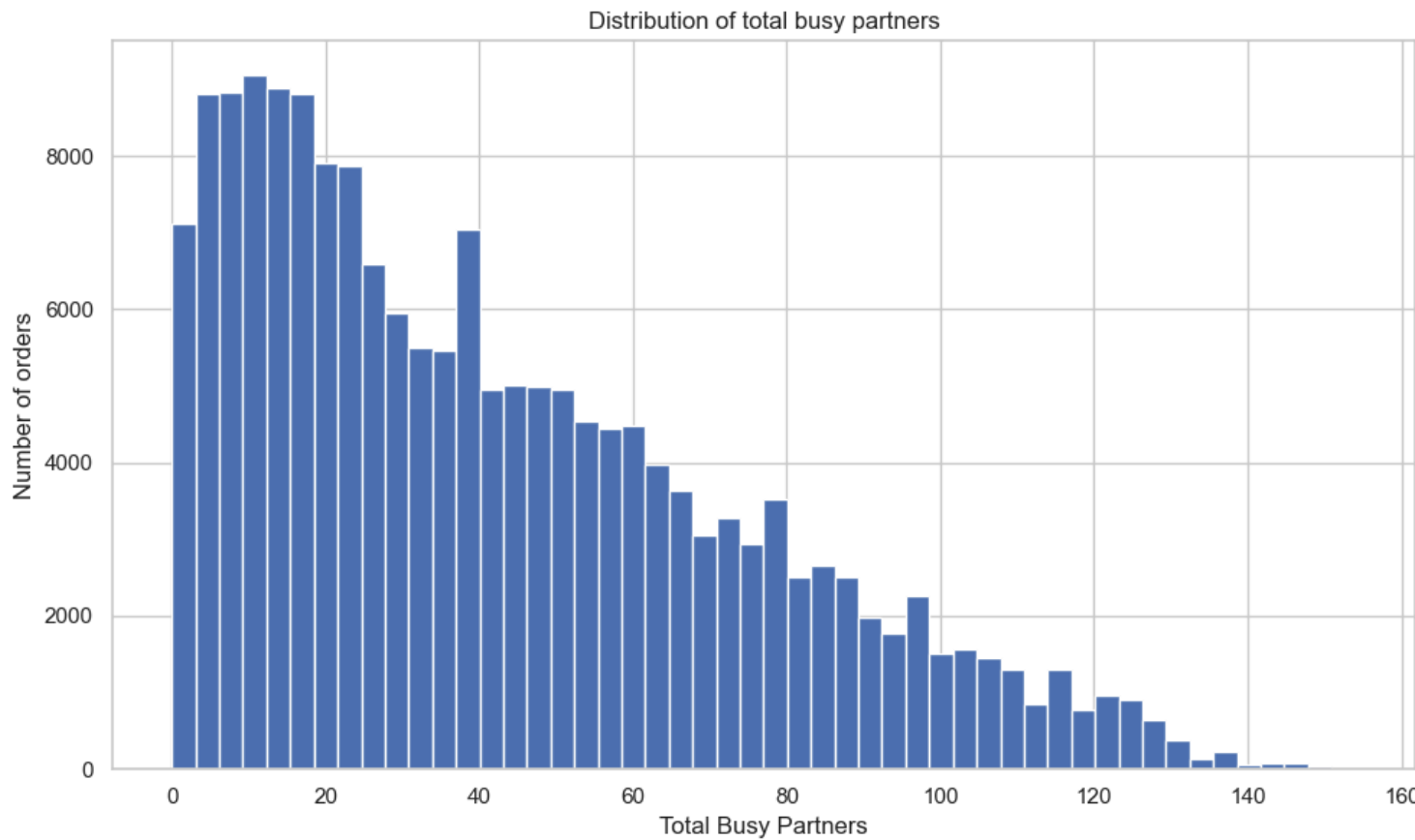
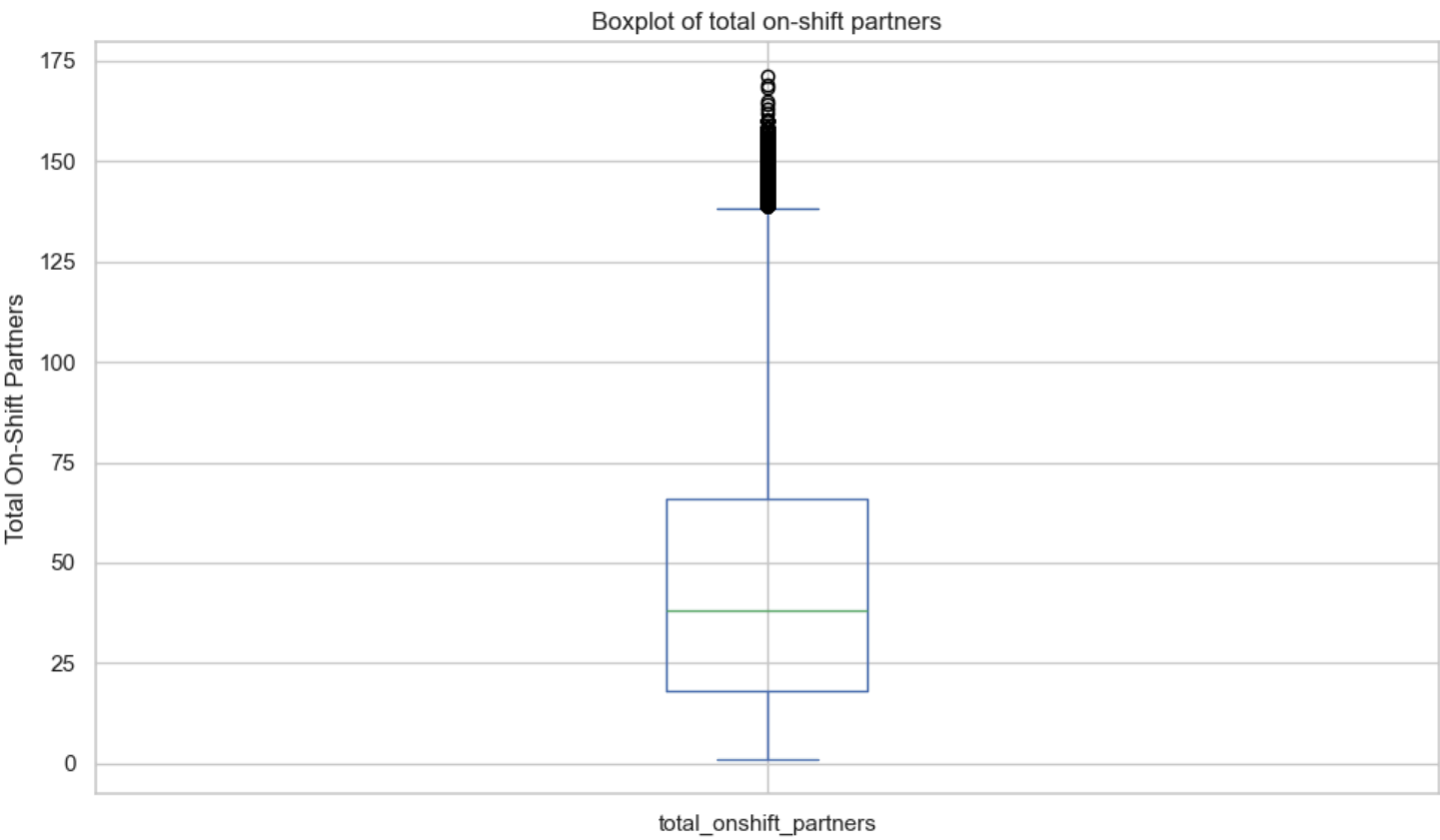
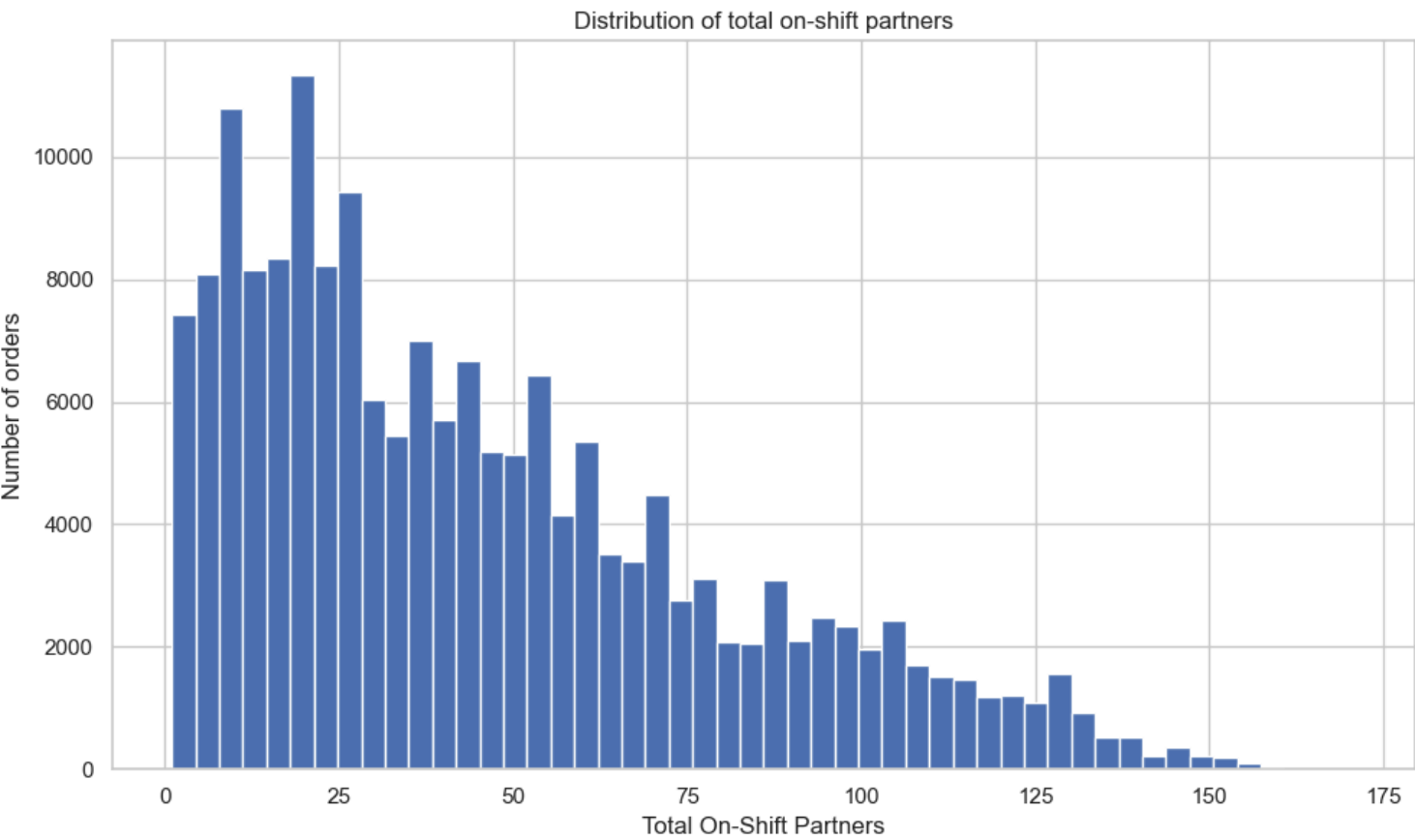
```
In [4]: fig, ax = plt.subplots(2,2, figsize=(20, 12))

df["total_onshift_partners"].hist(bins=50, ax=ax[0][0])
ax[0][0].set_title("Distribution of total on-shift partners")
ax[0][0].set_xlabel("Total On-Shift Partners")
ax[0][0].set_ylabel("Number of orders")

df["total_onshift_partners"].plot(kind="box", ax=ax[0][1])
ax[0][1].set_title("Boxplot of total on-shift partners")
ax[0][1].set_ylabel("Total On-Shift Partners")

# total_busy_partners
df["total_busy_partners"].hist(bins=50, ax=ax[1][0])
ax[1][0].set_title("Distribution of total busy partners")
ax[1][0].set_xlabel("Total Busy Partners")
ax[1][0].set_ylabel("Number of orders")
df["total_busy_partners"].plot(kind="box", ax=ax[1][1])
ax[1][1].set_title("Boxplot of total busy partners")
ax[1][1].set_ylabel("Total Busy Partners")

plt.tight_layout()
plt.show();
```

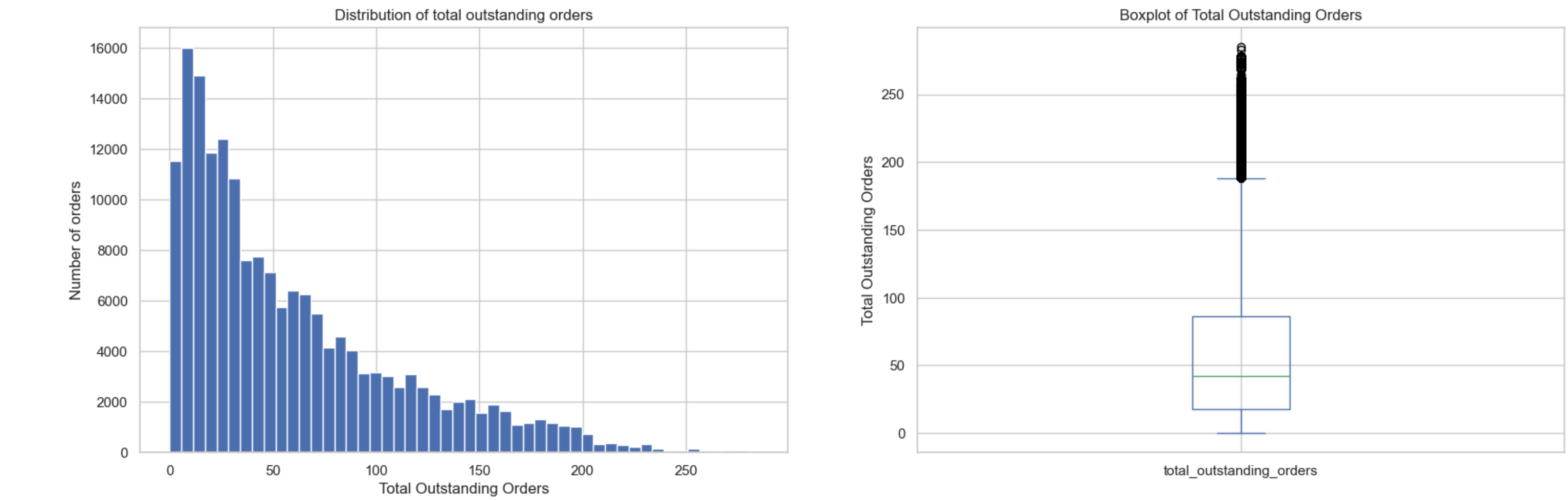


Observations

Total Outstanding Orders

```
In [9]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

df["total_outstanding_orders"].hist(bins=50, ax=ax[0])
ax[0].set_title("Distribution of total outstanding orders")
ax[0].set_xlabel("Total Outstanding Orders")
ax[0].set_ylabel("Number of orders")
df["total_outstanding_orders"].plot(kind="box", ax=ax[1])
ax[1].set_title("Boxplot of Total Outstanding Orders")
ax[1].set_ylabel("Total Outstanding Orders")
plt.show();
```

Observations

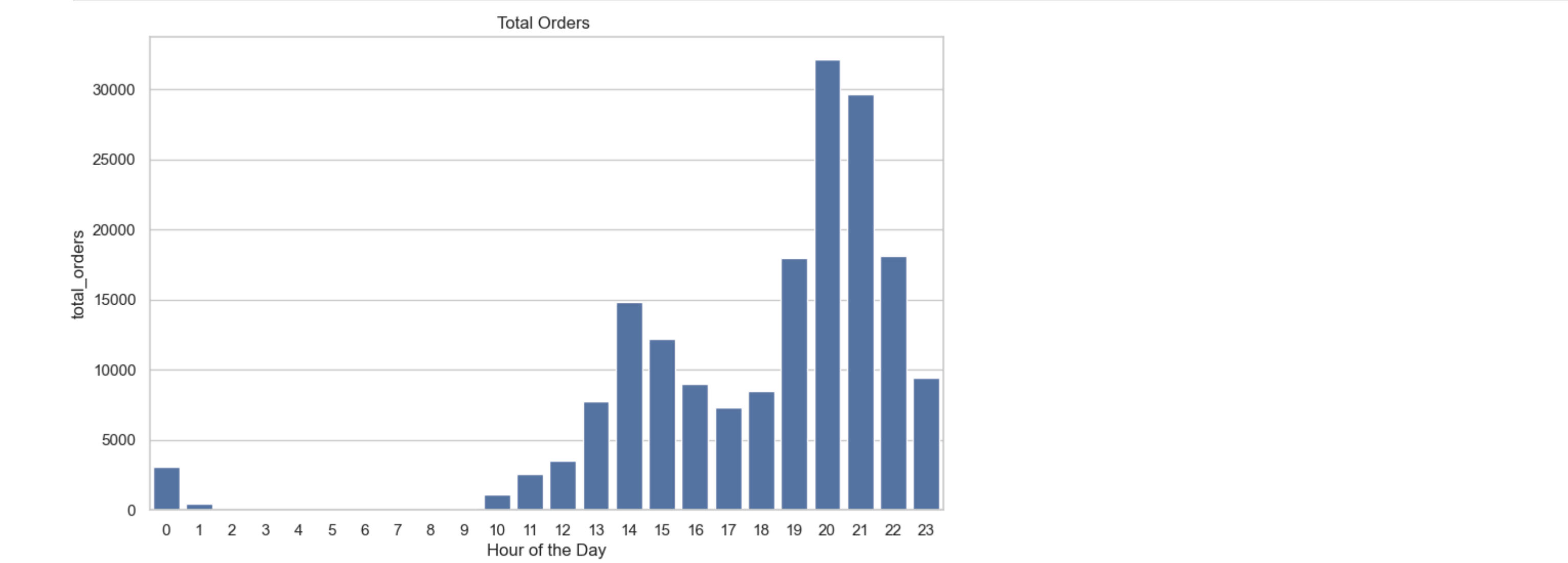
- At any given time there are around 30-80 outstanding orders

Order Created Hour

```
In [96]: full_hours = pd.DataFrame({'hour': range(0, 24)})

In [97]: full_hours=full_hours.copy().join(
df.groupby("created_at_hour").agg({"order_protocol": "size"}).rename(columns={"order_protocol": "total_orders"}),
on="hour",
how="left"
).fillna(0).set_index("hour")

In [98]: plt.figure(figsize=(10, 6))
sns.barplot(data=full_hours, x="hour", y="total_orders");
plt.title("Total Orders ")
plt.xlabel("Hour of the Day");
```



Observations

- We can see that during the lunch hours 1-2pm and dinner hours 7-10pm the number of orders is high

```
In [99]: orders_per_hour_df = db.sql("""
select created_at_hour as hour, round(count(*)/((date_diff('day', min(created_at), max(created_at))+1)*24),2) as orders_per_hour
from df
group by created_at_hour
order by created_at_hour
""").to_df()

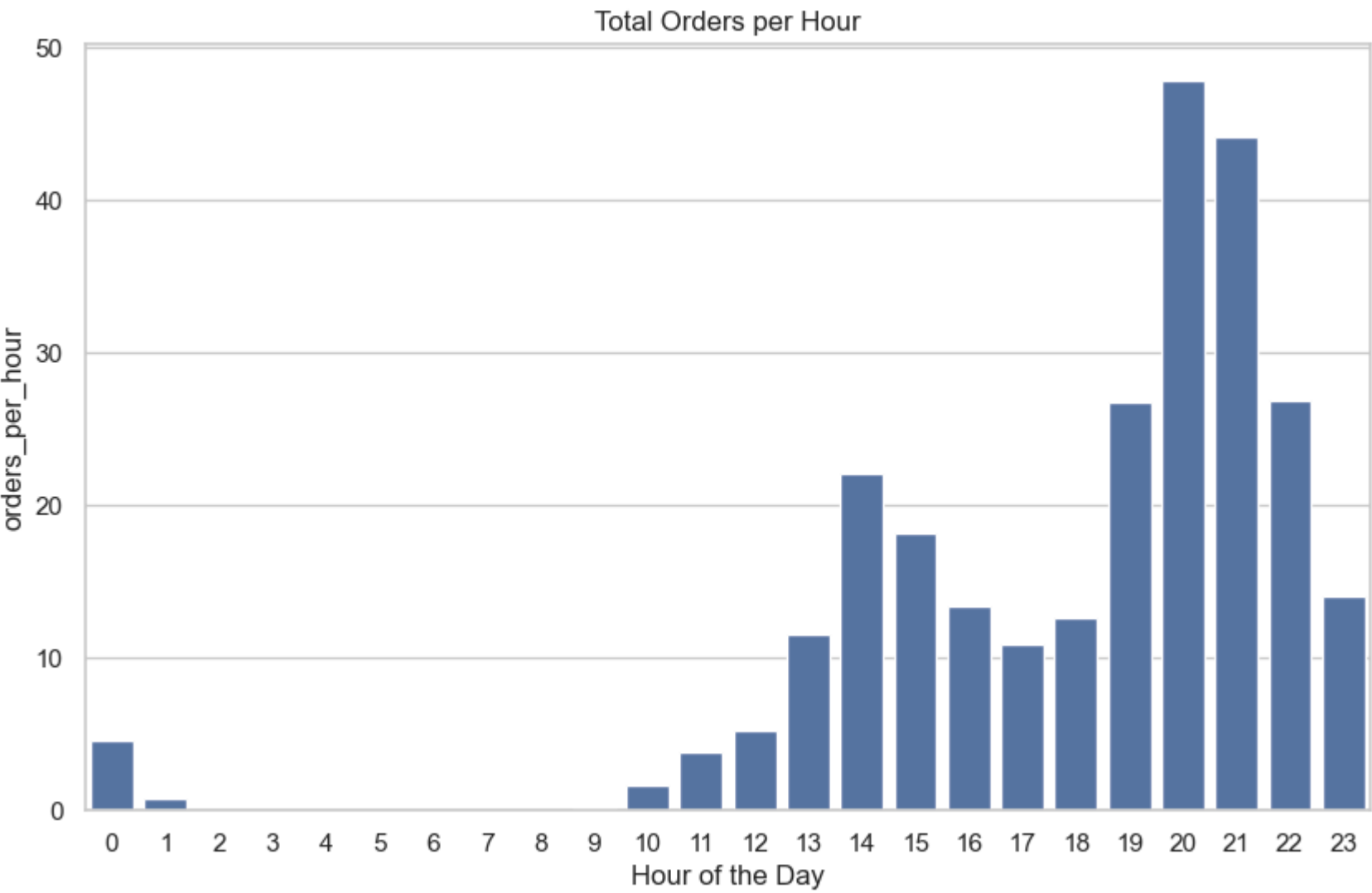
result_df = full_hours.copy().merge(orders_per_hour_df, on='hour', how='left').fillna(0)
result_df
```

Out [99]:

	hour	total_orders	orders_per_hour
0	0	3087.0	4.59
1	1	455.0	0.79
2	2	0.0	0.00
3	3	0.0	0.00
4	4	0.0	0.00
5	5	0.0	0.00
6	6	0.0	0.00
7	7	0.0	0.00
8	8	0.0	0.00
9	9	104.0	0.15
10	10	1075.0	1.60
11	11	2589.0	3.85
12	12	3484.0	5.18
13	13	7780.0	11.58
14	14	14810.0	22.04
15	15	12213.0	18.17
16	16	8993.0	13.38
17	17	7322.0	10.90
18	18	8459.0	12.59
19	19	17954.0	26.72
20	20	32117.0	47.79
21	21	29626.0	44.09
22	22	18080.0	26.90
23	23	9396.0	13.98

In [100]:

```
plt.figure(figsize=(10, 6))
sns.barplot(data=result_df, x="hour", y="orders_per_hour");
plt.title("Total Orders per Hour")
plt.xlabel("Hour of the Day");
```



Observations

- We can see that during the lunch hours 1-2pm and dinner hours 7-10pm the average number of orders is 25 and 45 respectively

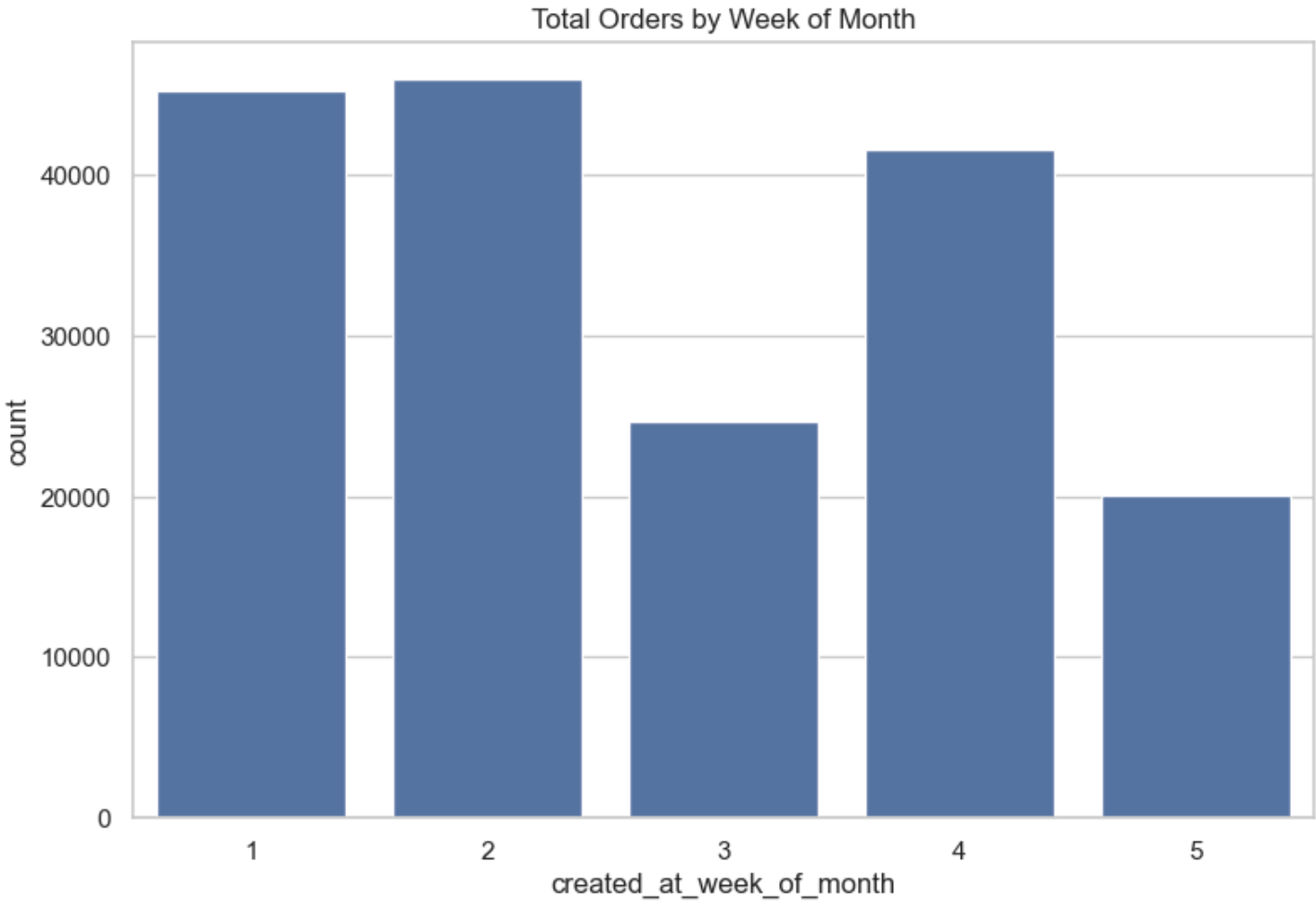
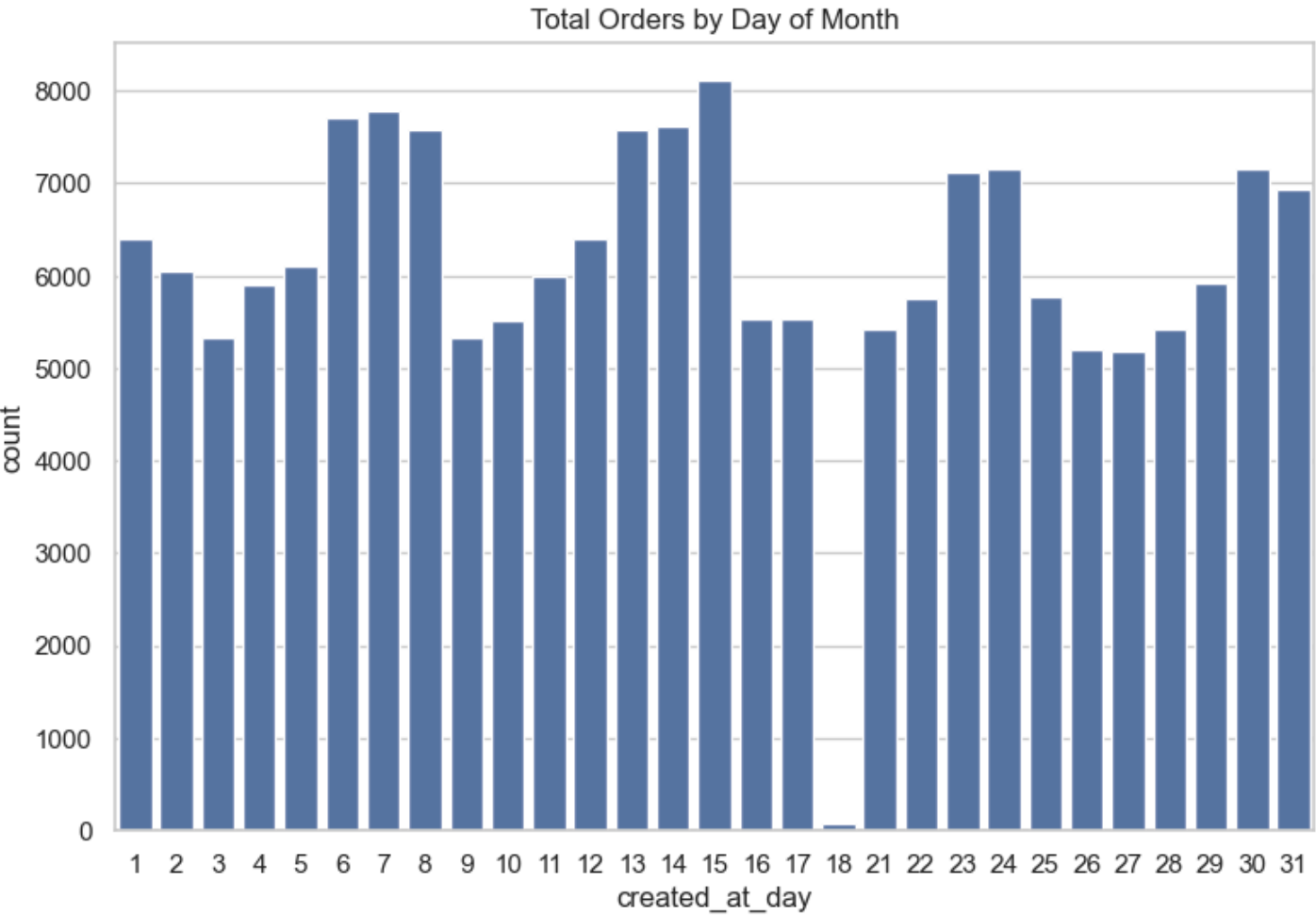
Order Created Week

In [9]:

```
fig, ax = plt.subplots(1,2, figsize=(20, 6))

sns.countplot(data=df, x="created_at_day", ax=ax[0]);
ax[0].set_title("Total Orders by Day of Month");

sns.countplot(data=df, x="created_at_week_of_month", ax=ax[1]);
ax[1].set_title("Total Orders by Week of Month");
plt.show();
```

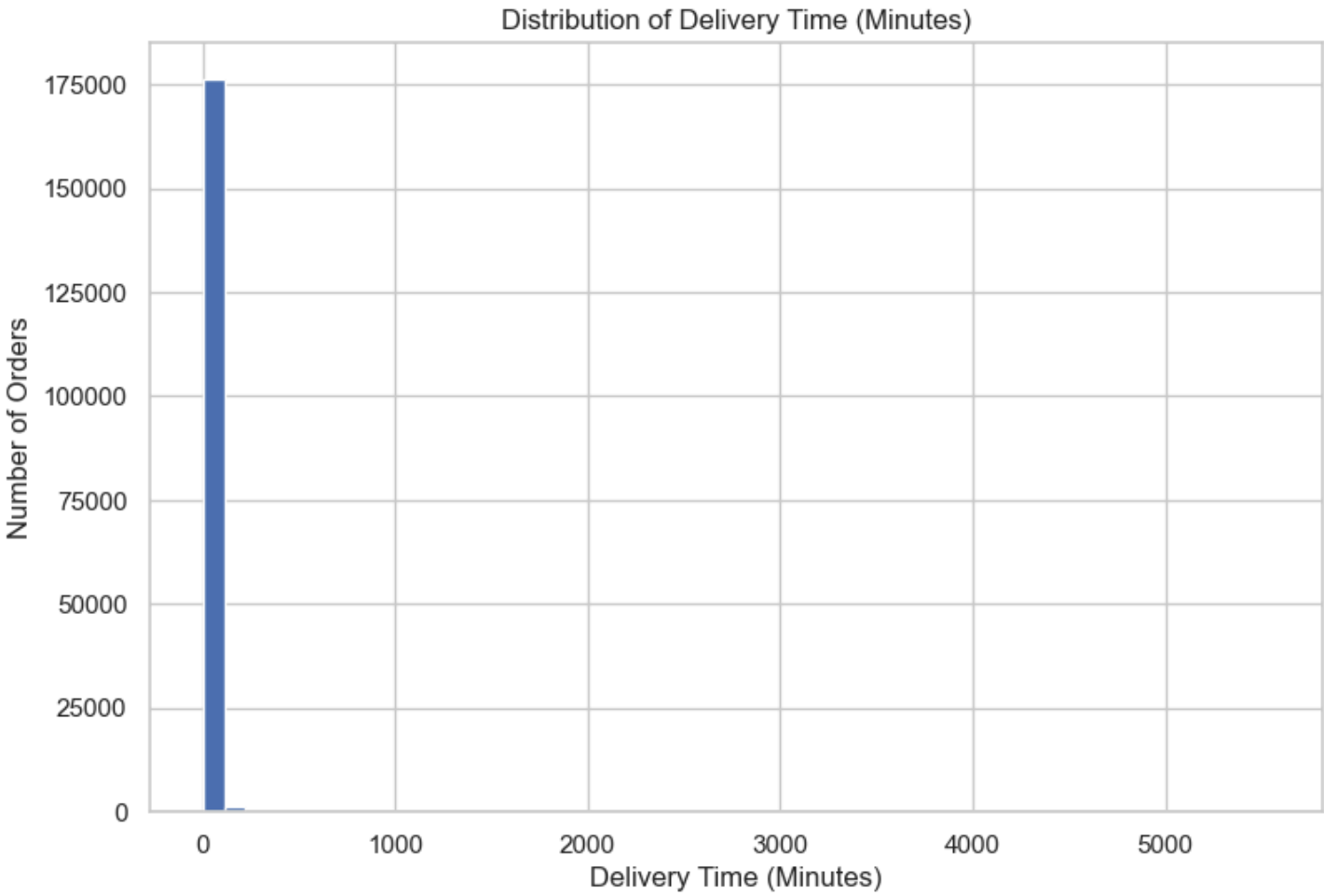


Observations

- We can see that there is a repeating pattern in the daily order data of a month
- We can see that orders are high in first 2 weeks and low in the last week of the month

```
In [11]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

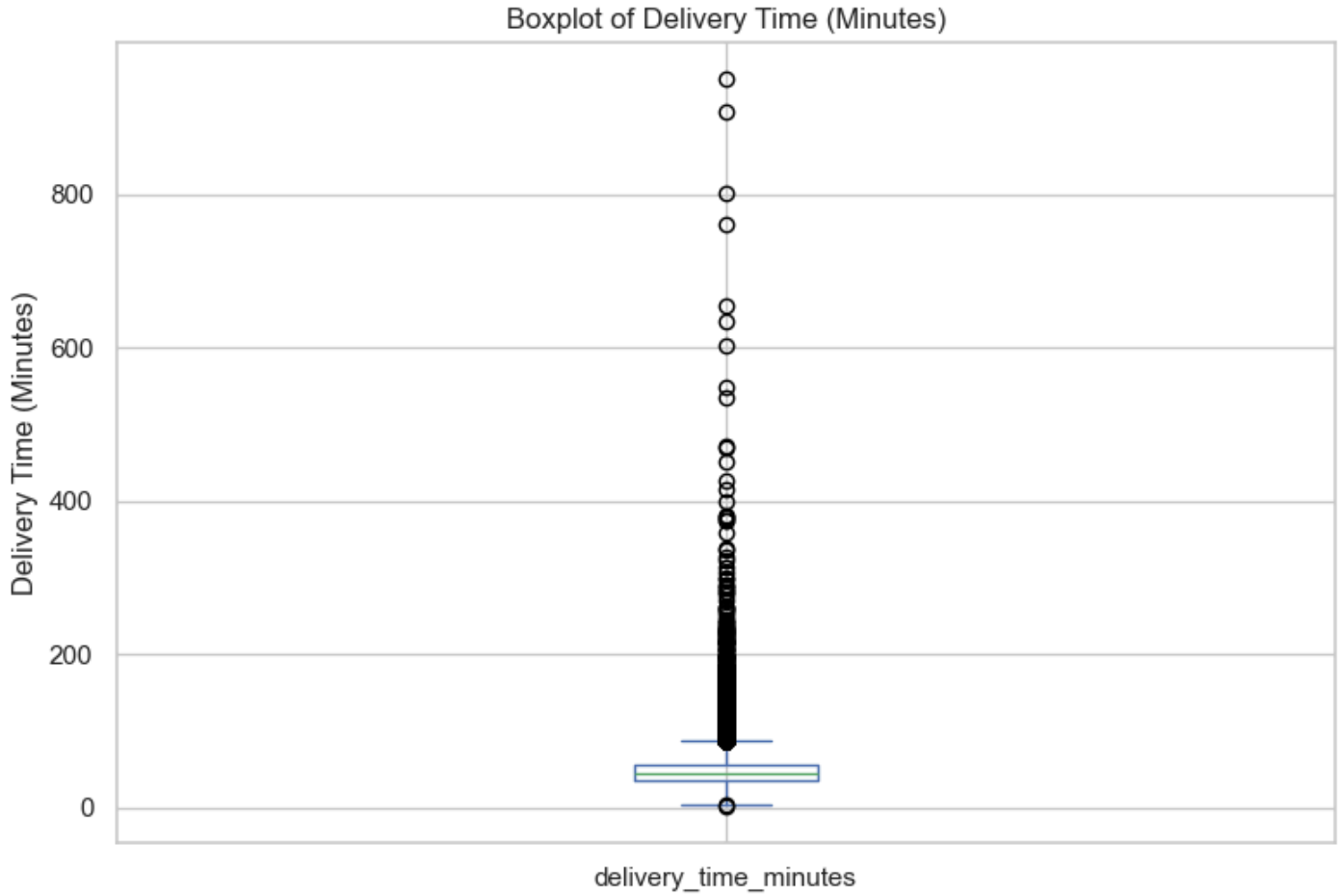
df["delivery_time_minutes"].hist(bins=50, ax=ax[0])
ax[0].set_title("Distribution of Delivery Time (Minutes)")
ax[0].set_xlabel("Delivery Time (Minutes)")
ax[0].set_ylabel("Number of Orders")
df["delivery_time_minutes"].plot(kind="box", ax=ax[1])
ax[1].set_title("Boxplot of Delivery Time (Minutes)")
ax[1].set_ylabel("Delivery Time (Minutes)")
plt.show()
```



```
In [13]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

_=df[df["delivery_time_minutes"]<1000]

_["delivery_time_minutes"].hist(bins=100, ax=ax[0])
ax[0].set_title("Distribution of Delivery Time (Minutes)")
ax[0].set_xlabel("Delivery Time (Minutes)")
ax[0].set_ylabel("Number of Orders")
_["delivery_time_minutes"].plot(kind="box", ax=ax[1])
ax[1].set_title("Boxplot of Delivery Time (Minutes)")
ax[1].set_ylabel("Delivery Time (Minutes)")
plt.show()
```



Observations

- Most of the delivery times are under 100 minutes

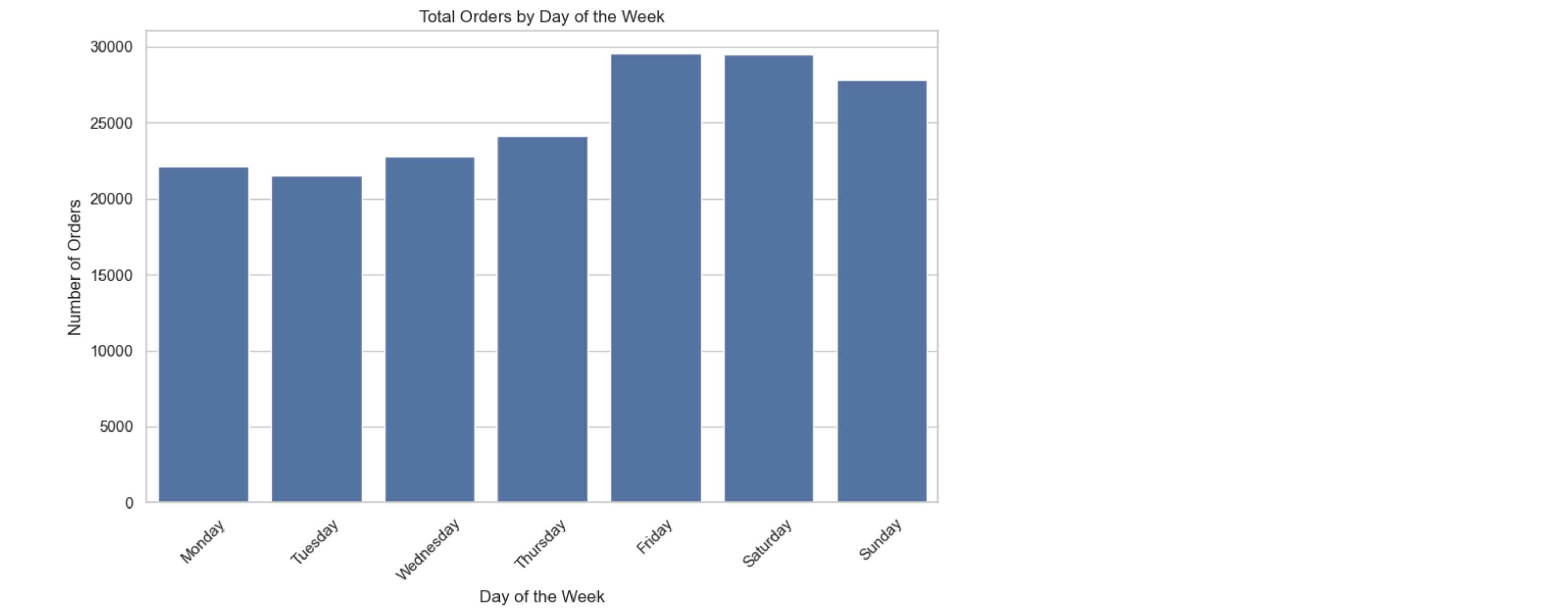
```
In [37]: temp=df["created_at_day_of_week"].value_counts().reset_index().sort_values(by="created_at_day_of_week")
temp
```

Out[37]:

	created_at_day_of_week	count
5	1	22105
6	2	21547
4	3	22796
3	4	24177
0	5	29562
1	6	29490
2	7	27867

```
In [42]: plt.figure(figsize=(10, 6))

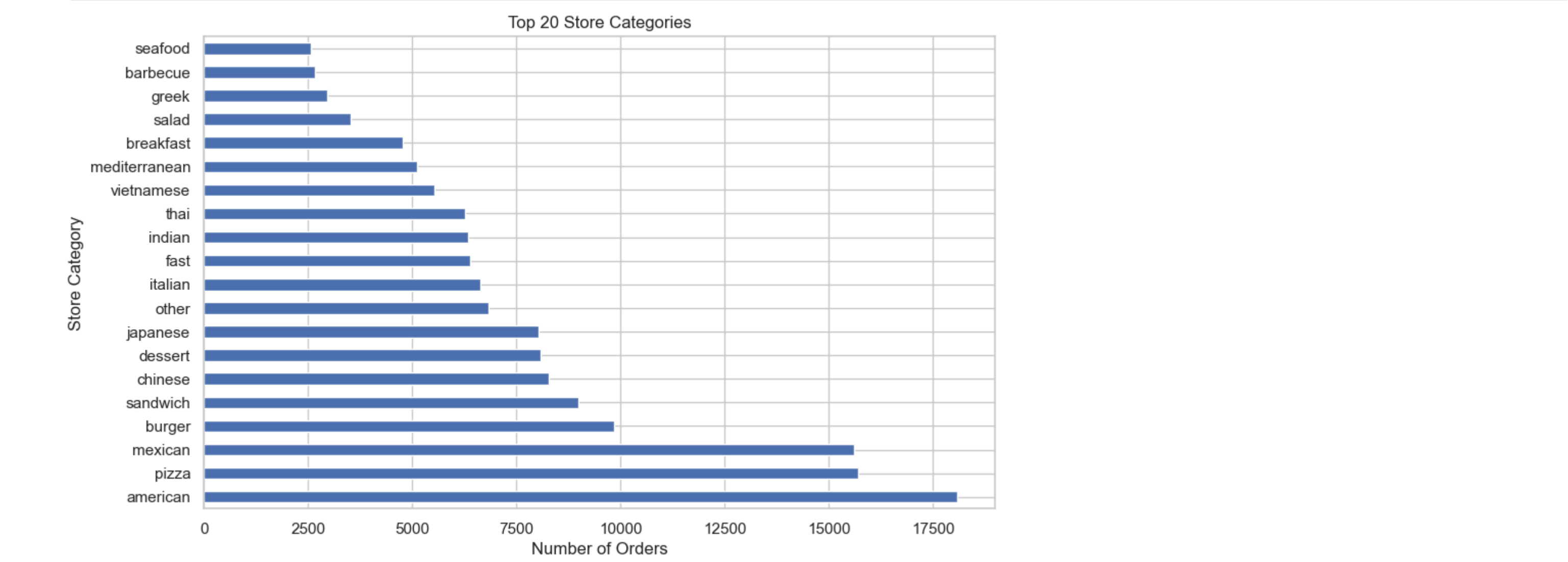
sns.barplot(data=temp, x="created_at_day_of_week", y="count")
day_names = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
tick_positions = plt.xticks()[0]
plt.xticks(tick_positions, day_names, rotation=45)
plt.xlabel("Day of the Week")
plt.ylabel("Number of Orders")
plt.title("Total Orders by Day of the Week");
```

Observations

- Most of the deliveries happen on Friday, Saturday and Sunday

```
In [71]: df["store_primary_category"].value_counts().head(20).plot(kind="barh", figsize=(10, 6));
plt.title("Top 20 Store Categories")
plt.xlabel("Number of Orders")
plt.ylabel("Store Category")
plt.show();
```



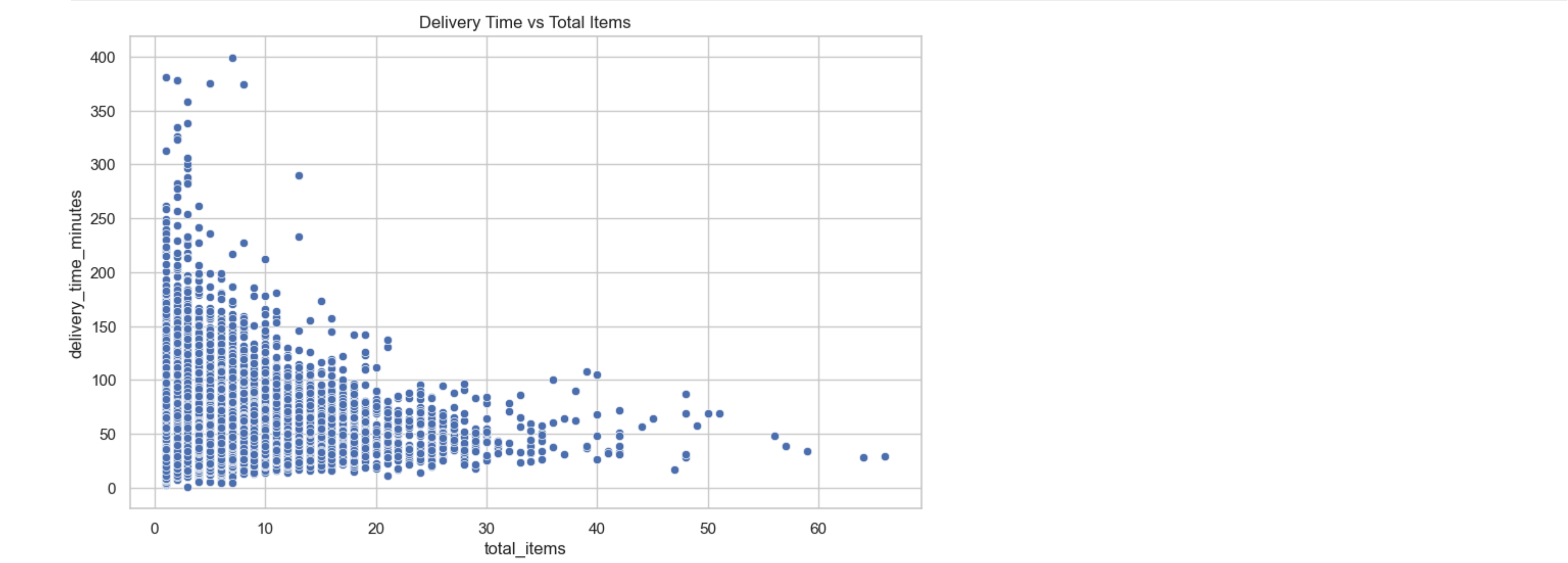
Observations

- We can see that American, Pizza, and Mexican are the most ordered categories

Bivariate Analysis

```
In [8]: t_df=df[(df["delivery_time_minutes"]<400) & (df["total_items"]<100) & (df["max_item_price"]<10000)]
```

```
In [114]: plt.figure(figsize=(10, 6))
sns.scatterplot(data=t_df, x='total_items', y='delivery_time_minutes');
plt.title("Delivery Time vs Total Items");
```

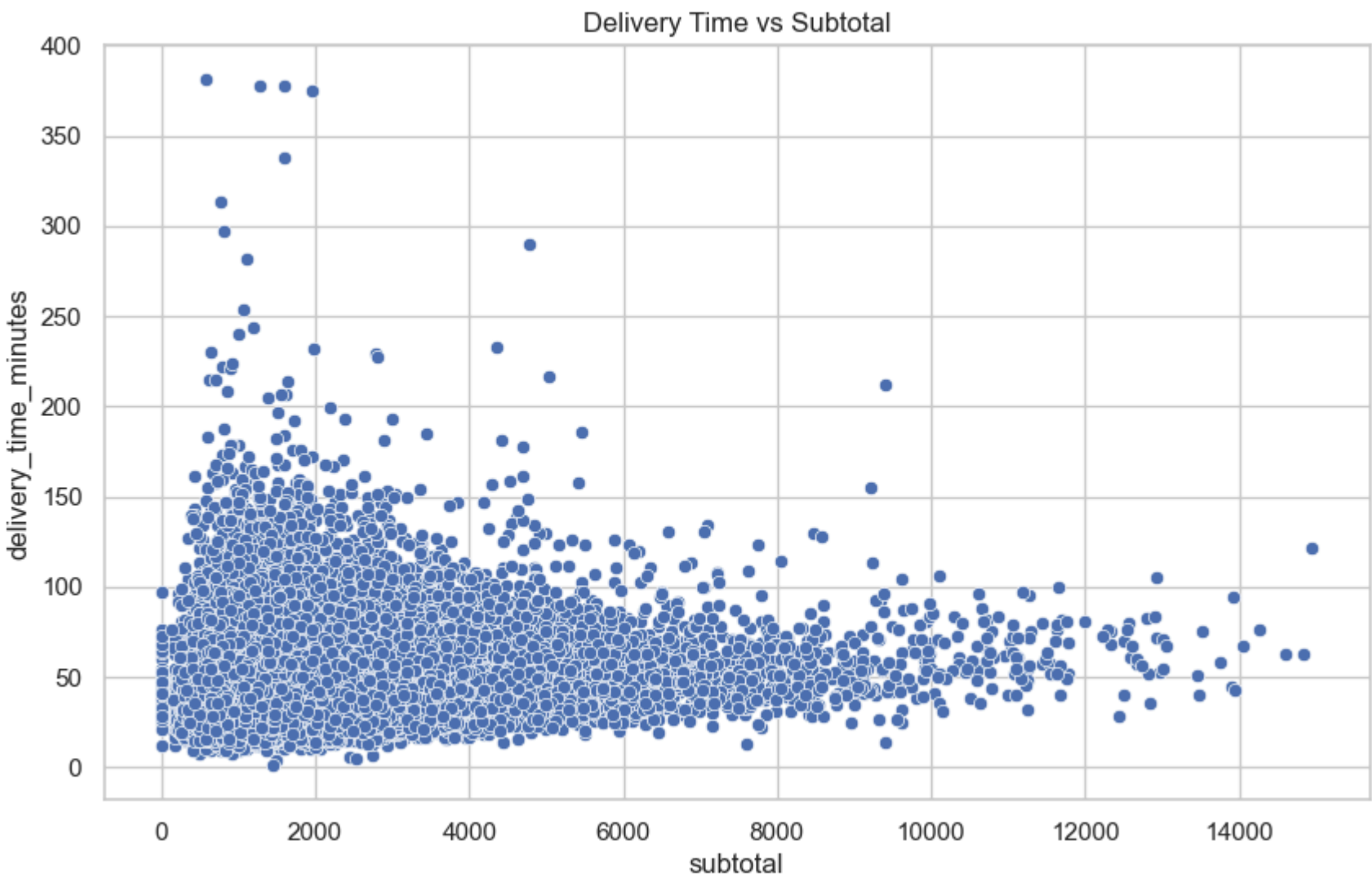


Observations

- We can see that delivery time decreases as the number of total items increases

Subtotal vs Total Items

```
In [99]: plt.figure(figsize=(10, 6))
sns.scatterplot(data=t_df, x='subtotal', y='delivery_time_minutes');
plt.title("Delivery Time vs Subtotal");
```

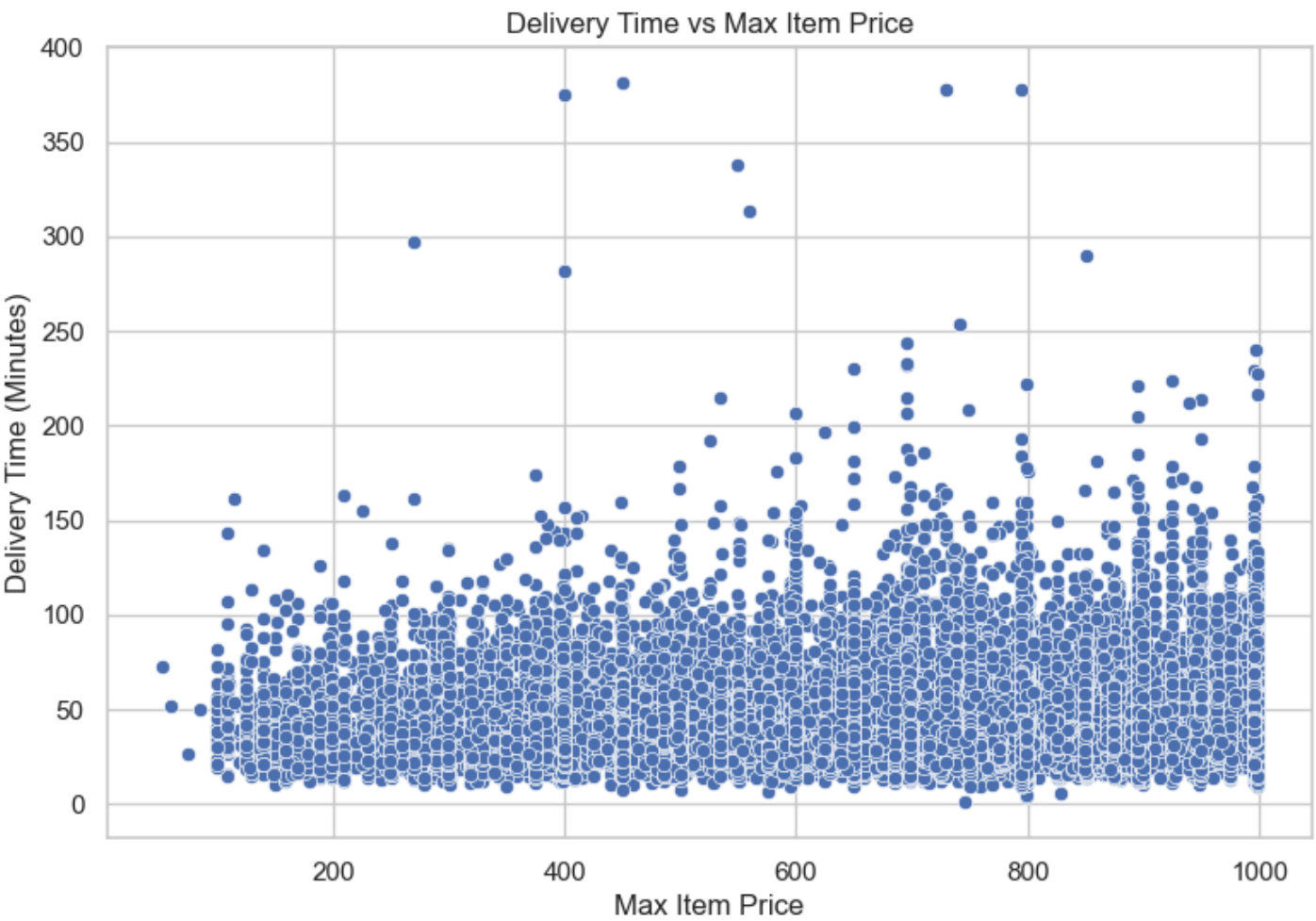
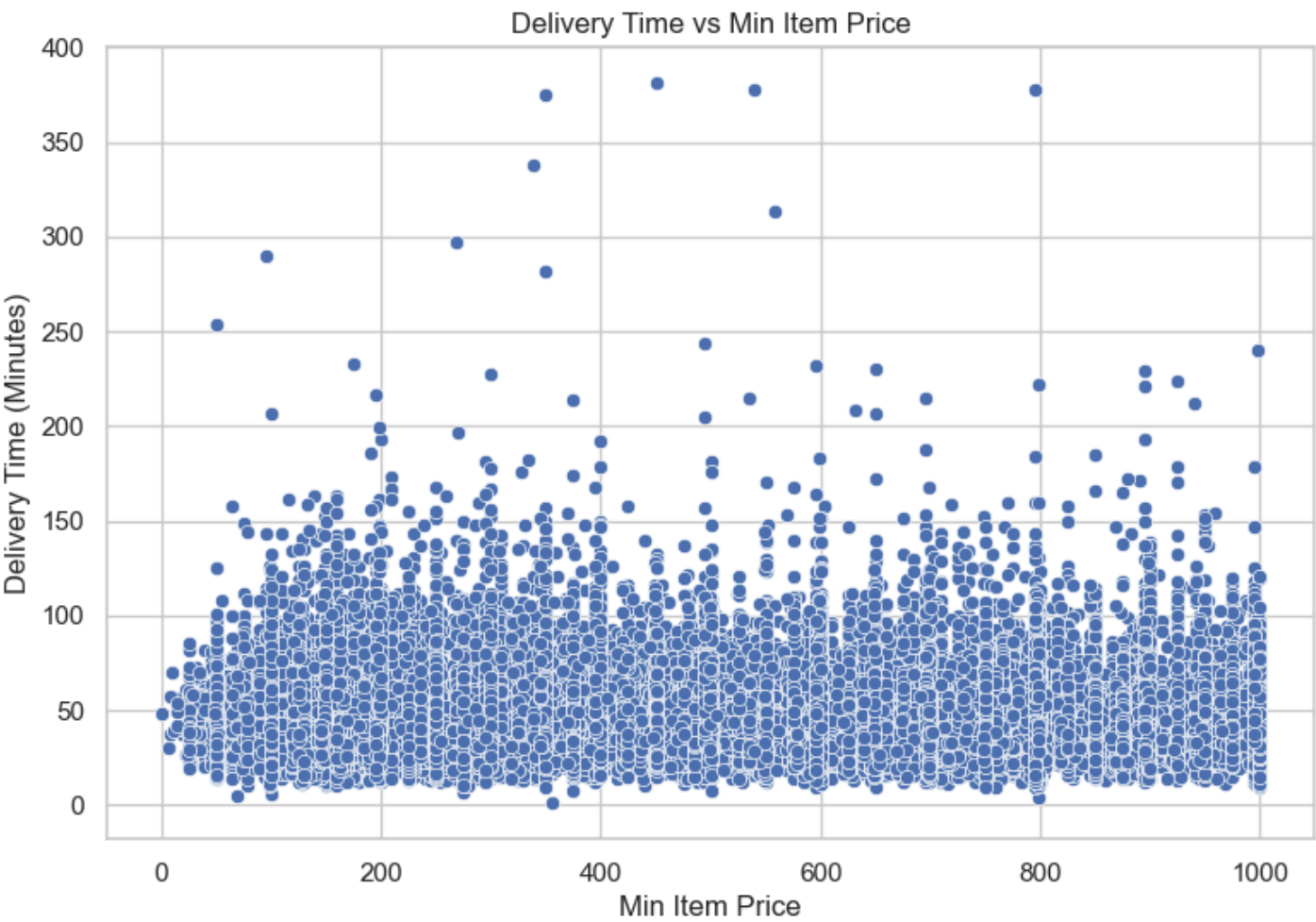


Observations

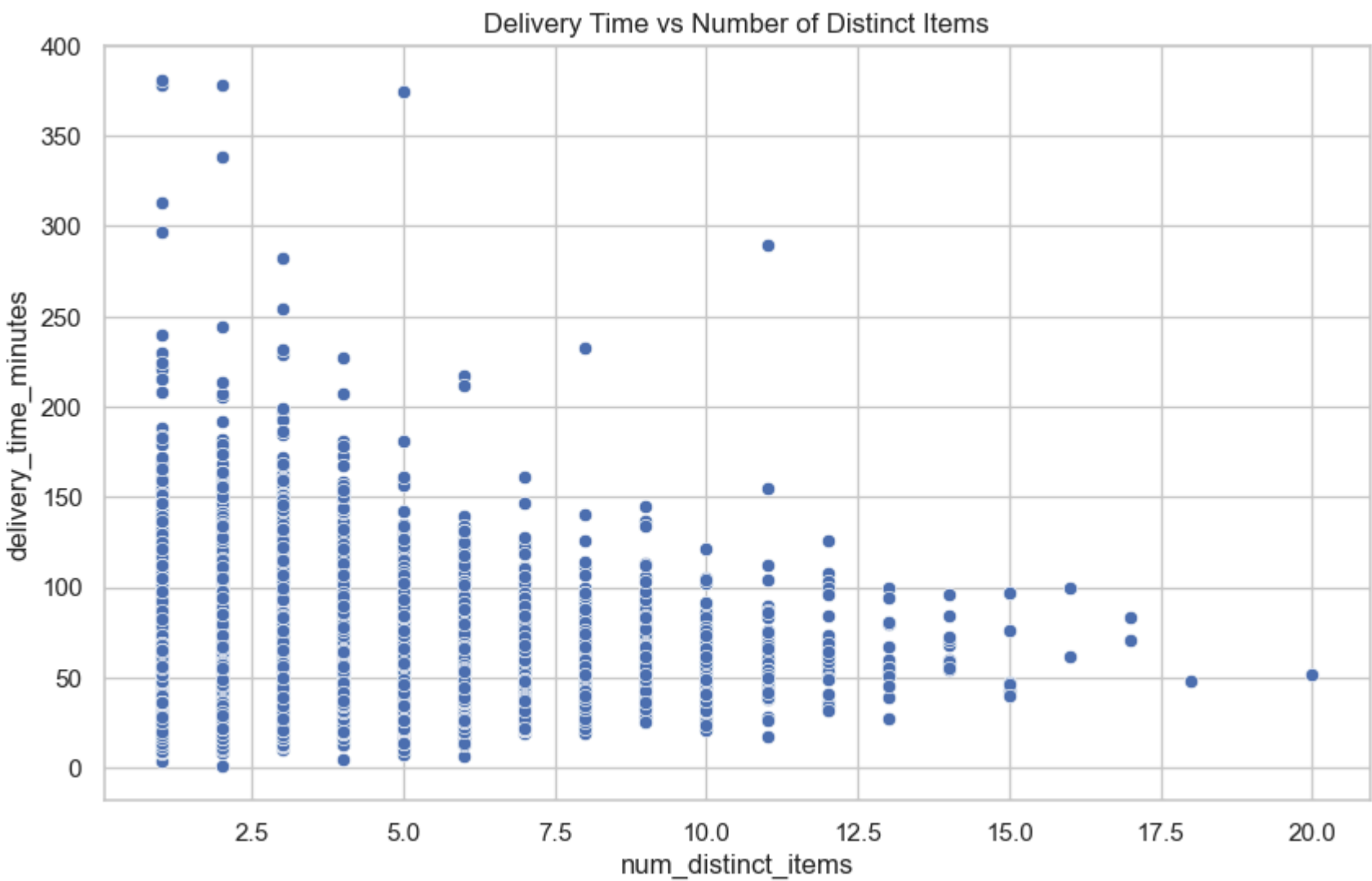
```
In [100]: fig, ax = plt.subplots(1,2, figsize=(20, 6))

sns.scatterplot(data=t_df, x='min_item_price', y='delivery_time_minutes', ax=ax[0]);
ax[0].set_title("Delivery Time vs Min Item Price");
ax[0].set_xlabel("Min Item Price")
ax[0].set_ylabel("Delivery Time (Minutes)")

sns.scatterplot(data=t_df, x='max_item_price', y='delivery_time_minutes', ax=ax[1]);
ax[1].set_title("Delivery Time vs Max Item Price");
ax[1].set_xlabel("Max Item Price")
ax[1].set_ylabel("Delivery Time (Minutes)");
```



```
In [101]: plt.figure(figsize=(10, 6))
sns.scatterplot(data=t_df, x='num_distinct_items', y='delivery_time_minutes');
plt.title("Delivery Time vs Number of Distinct Items");
```



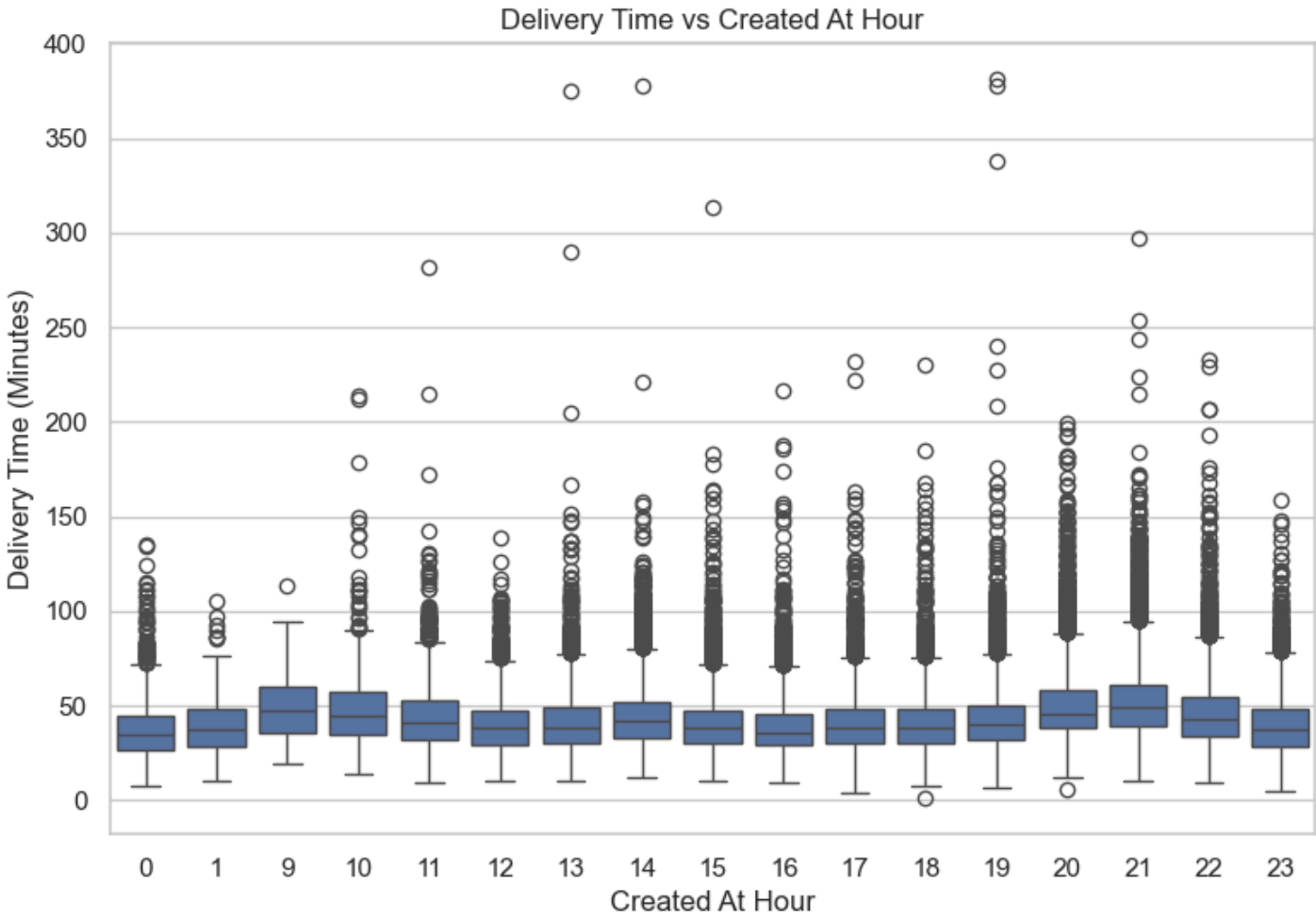
Observations

- We can see that the delivery time is less as the number of unique items increases


```
In [102... fig, ax = plt.subplots(1,2, figsize=(20, 6))

sns.scatterplot(data=t_df, x='created_at_hour', y='delivery_time_minutes', ax=ax[0]);
ax[0].set_title("Delivery Time vs Created At Hour");
ax[0].set_xlabel("Created At Hour")
ax[0].set_ylabel("Delivery Time (Minutes)")

sns.boxplot(data=t_df, x='created_at_hour', y='delivery_time_minutes', ax=ax[1]);
ax[1].set_title("Delivery Time vs Created At Hour");
ax[1].set_xlabel("Created At Hour");
ax[1].set_ylabel("Delivery Time (Minutes)");
```



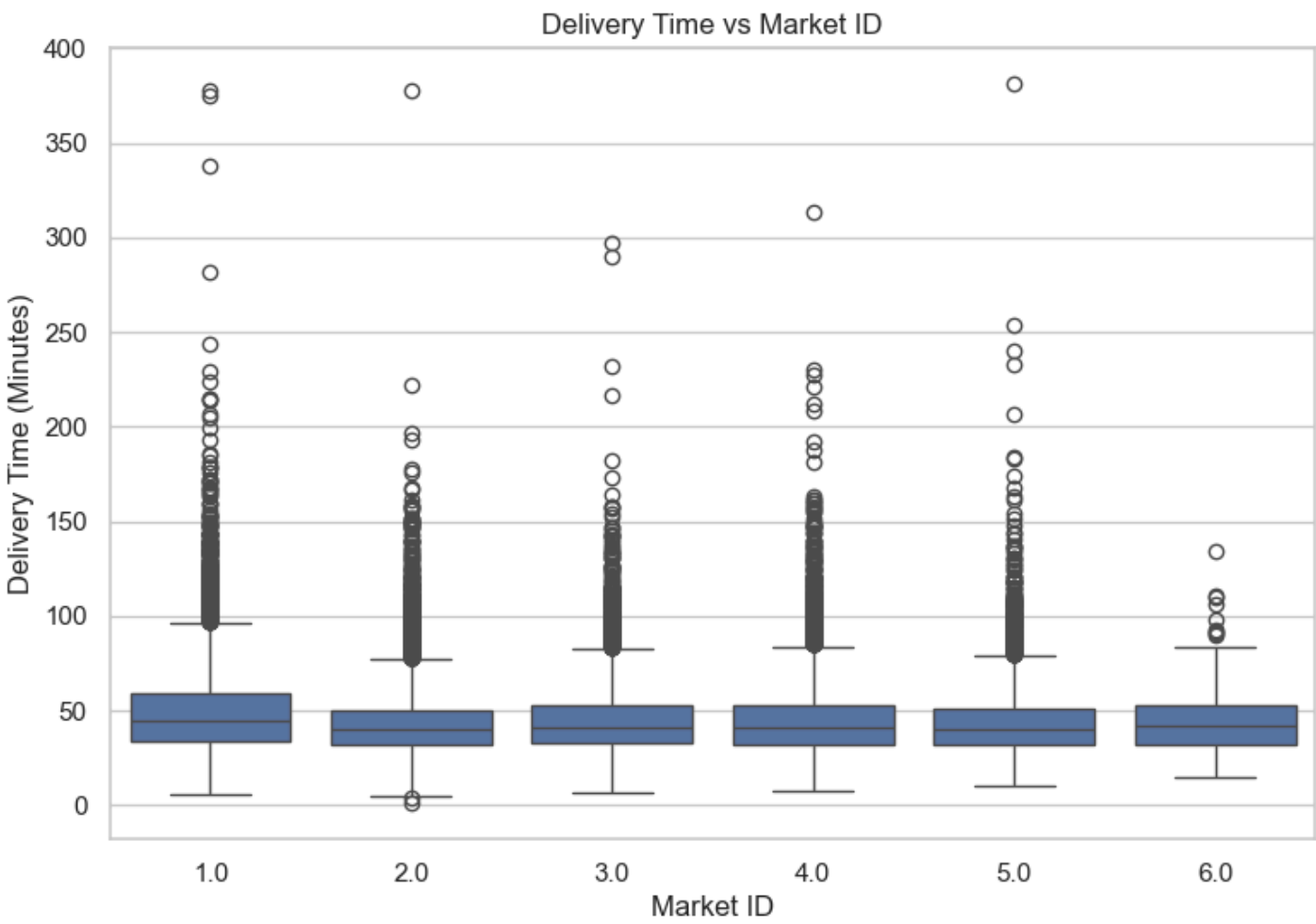
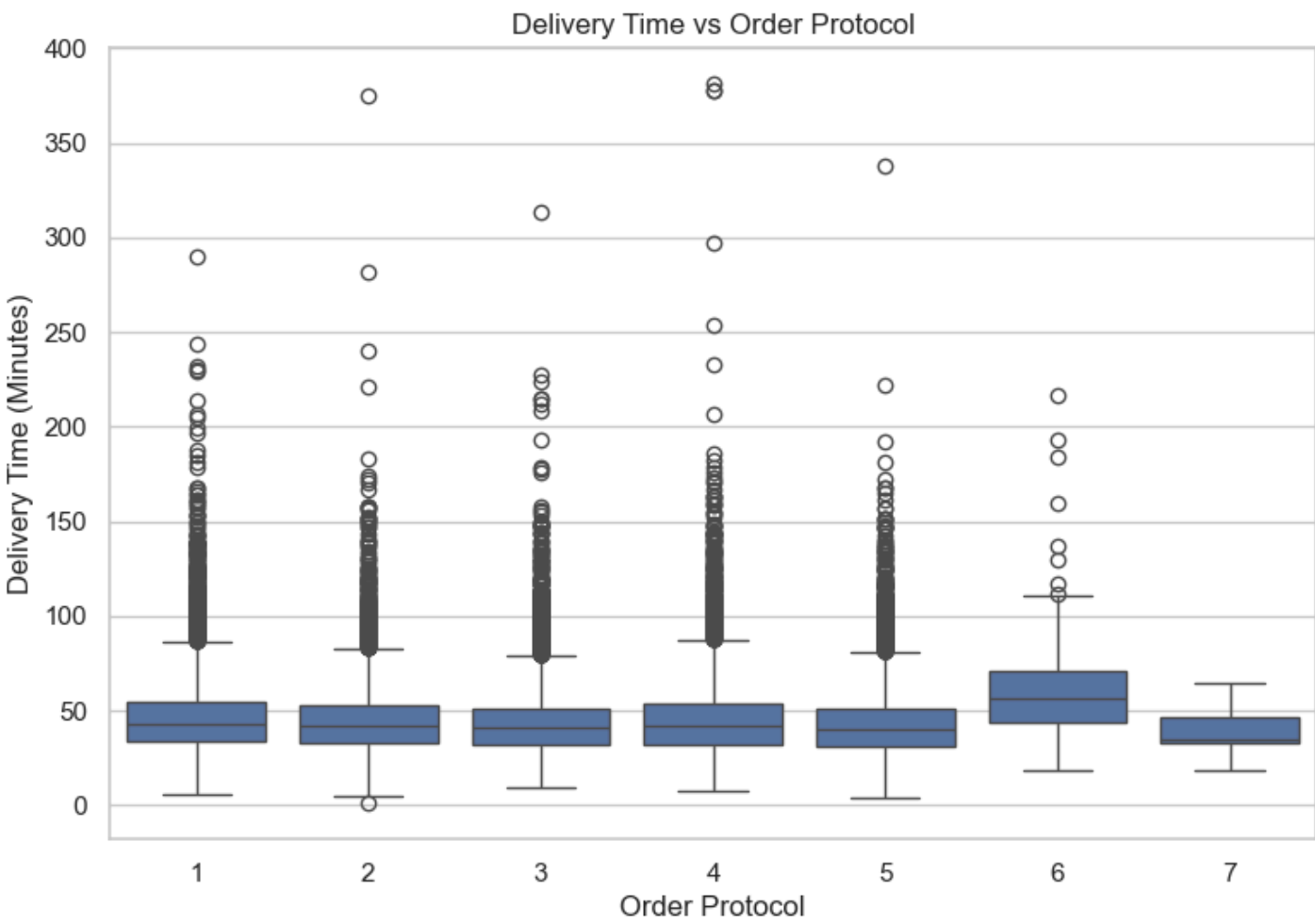
Observations

- We can see that delivery time increases during peak hours

```
In [103... fig, ax = plt.subplots(1,2, figsize=(20, 6))

sns.boxplot(data=t_df, x='order_protocol', y='delivery_time_minutes', ax=ax[0]);
ax[0].set_title("Delivery Time vs Order Protocol");
ax[0].set_xlabel("Order Protocol");
ax[0].set_ylabel("Delivery Time (Minutes)");

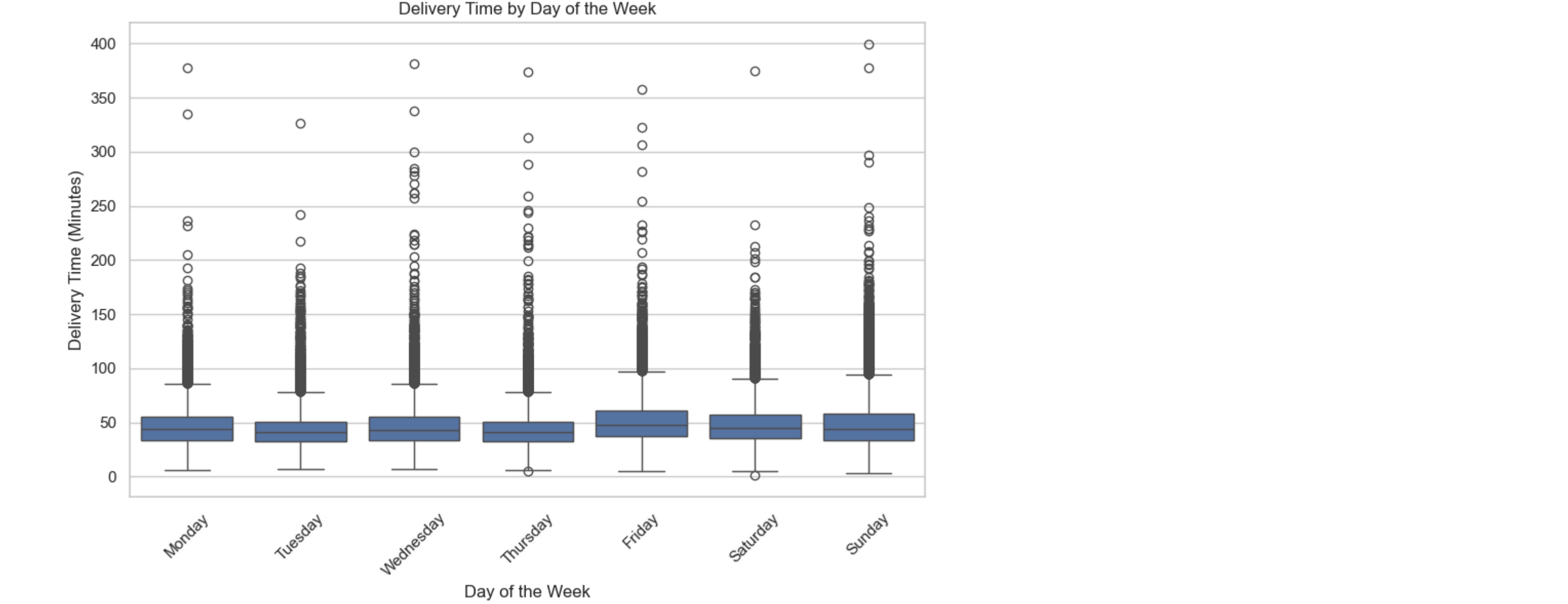
sns.boxplot(data=t_df, x='market_id', y='delivery_time_minutes', ax=ax[1]);
ax[1].set_title("Delivery Time vs Market ID");
ax[1].set_xlabel("Market ID");
ax[1].set_ylabel("Delivery Time (Minutes)");
```



Observations

- Order protocol 6 had a slightly higher delivery time

```
In [10]: plt.figure(figsize=(10, 6))
sns.boxplot(data=t_df, x="created_at_day_of_week", y="delivery_time_minutes")
plt.title("Delivery Time by Day of the Week")
plt.xlabel("Day of the Week")
plt.ylabel("Delivery Time (Minutes)")
day_names = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
tick_positions = plt.xticks()[0]
plt.xticks(tick_positions, day_names, rotation=45)
plt.show()
```

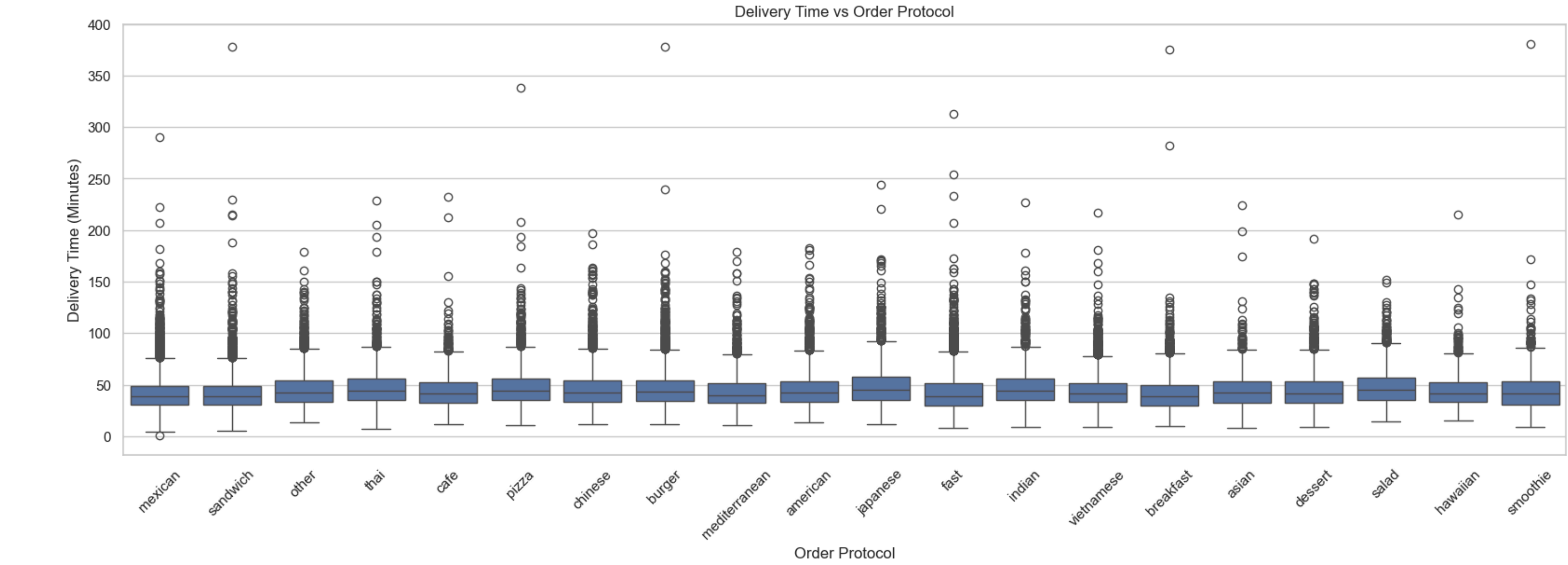


```
In [104... top_cat=t_df["store_primary_category"].value_counts().head(20).index
top_cat
t_df[t_df["store_primary_category"].isin(top_cat)]
```

	market_id	created_at	actual_delivery_time	store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	...	actual_delivery_time
19	1.0	2015-01-30 20:51:23	2015-01-30 21:41:42	ce016f59ecc2366a43e1c96a4774d167	mexican	1	2	950	2	150.0	...	
20	1.0	2015-01-31 18:15:12	2015-01-31 18:44:05	ce016f59ecc2366a43e1c96a4774d167	mexican	1	5	1285	3	150.0	...	
31	3.0	2015-01-30 14:19:30	2015-01-30 14:58:45	50905d7b2216bfeccb5b41016357176b	sandwich	2	4	3490	3	175.0	...	
32	1.0	2015-01-28 15:03:04	2015-01-28 15:34:14	50905d7b2216bfeccb5b41016357176b	other	2	3	1765	3	275.0	...	
33	1.0	2015-02-12 14:57:42	2015-02-12 15:48:29	50905d7b2216bfeccb5b41016357176b	other	2	1	975	1	975.0	...	
...	...	...	...	...	...	...	...	...	...	...	...	
177539	1.0	2015-02-16 18:49:41	2015-02-16 19:54:48	a914ecef9c12ffdb9bede64bb703d877	fast	4	3	1389	3	345.0	...	
177540	1.0	2015-02-12 18:31:59	2015-02-12 19:28:22	a914ecef9c12ffdb9bede64bb703d877	fast	4	6	3010	4	405.0	...	
177541	1.0	2015-01-23 23:16:08	2015-01-24 00:06:16	a914ecef9c12ffdb9bede64bb703d877	fast	4	5	1836	3	300.0	...	
177542	1.0	2015-02-01 12:48:15	2015-02-01 13:53:22	c81e155d85dae5430a8cee6f2242e82c	sandwich	1	1	1175	1	535.0	...	
177543	1.0	2015-02-08 13:54:33	2015-02-08 14:31:41	c81e155d85dae5430a8cee6f2242e82c	sandwich	1	4	2605	4	425.0	...	

70721 rows x 39 columns

```
In [ ]: plt.figure(figsize=(20, 6))
sns.boxplot(data=t_df[t_df["store_primary_category"].isin(top_cat)], x='store_primary_category', y='delivery_time_minutes')
plt.title("Delivery Time vs Store primary category")
plt.xlabel("Store primary category")
plt.ylabel("Delivery Time (Minutes)")
plt.xticks(rotation=45);
```



Observations

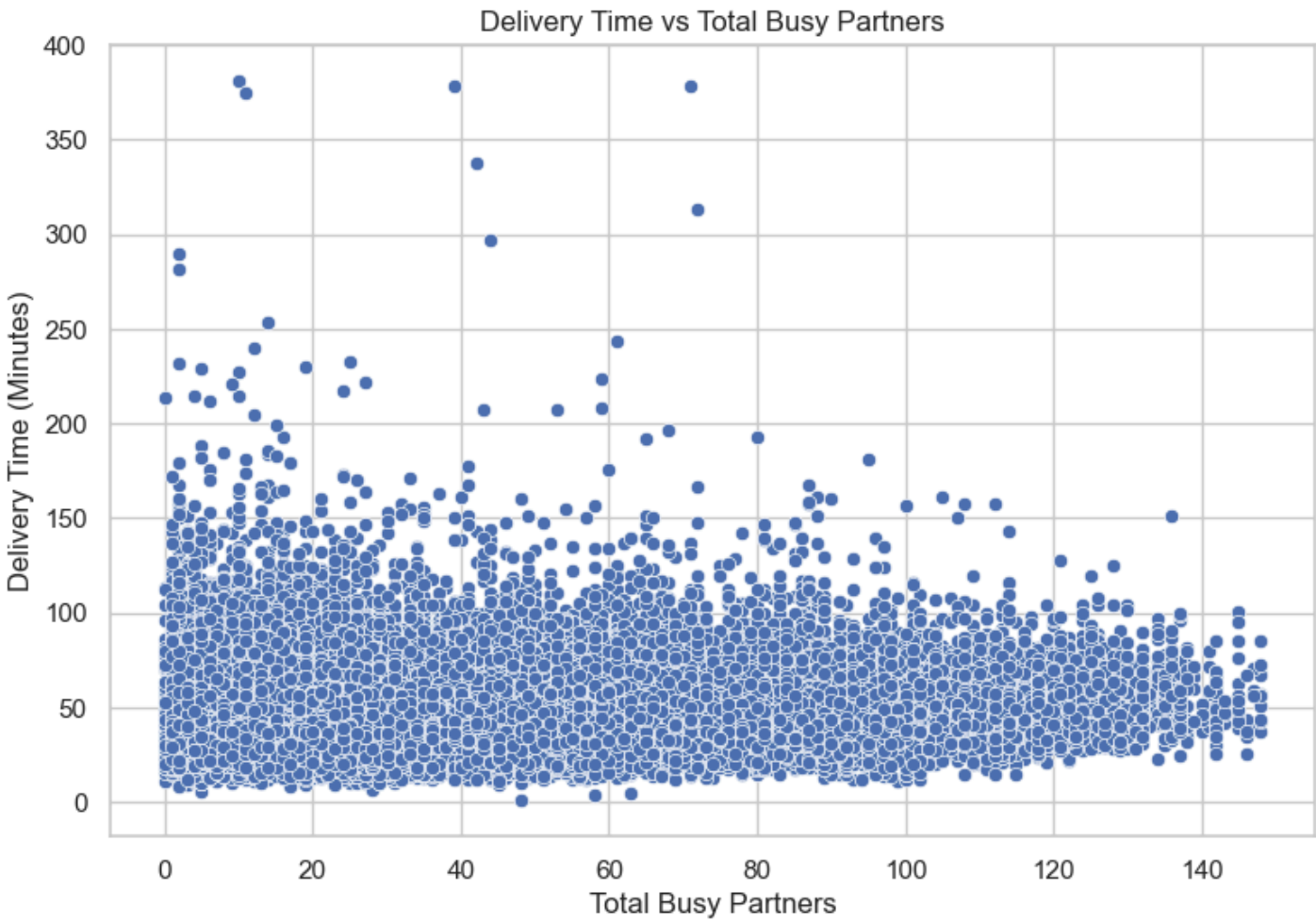
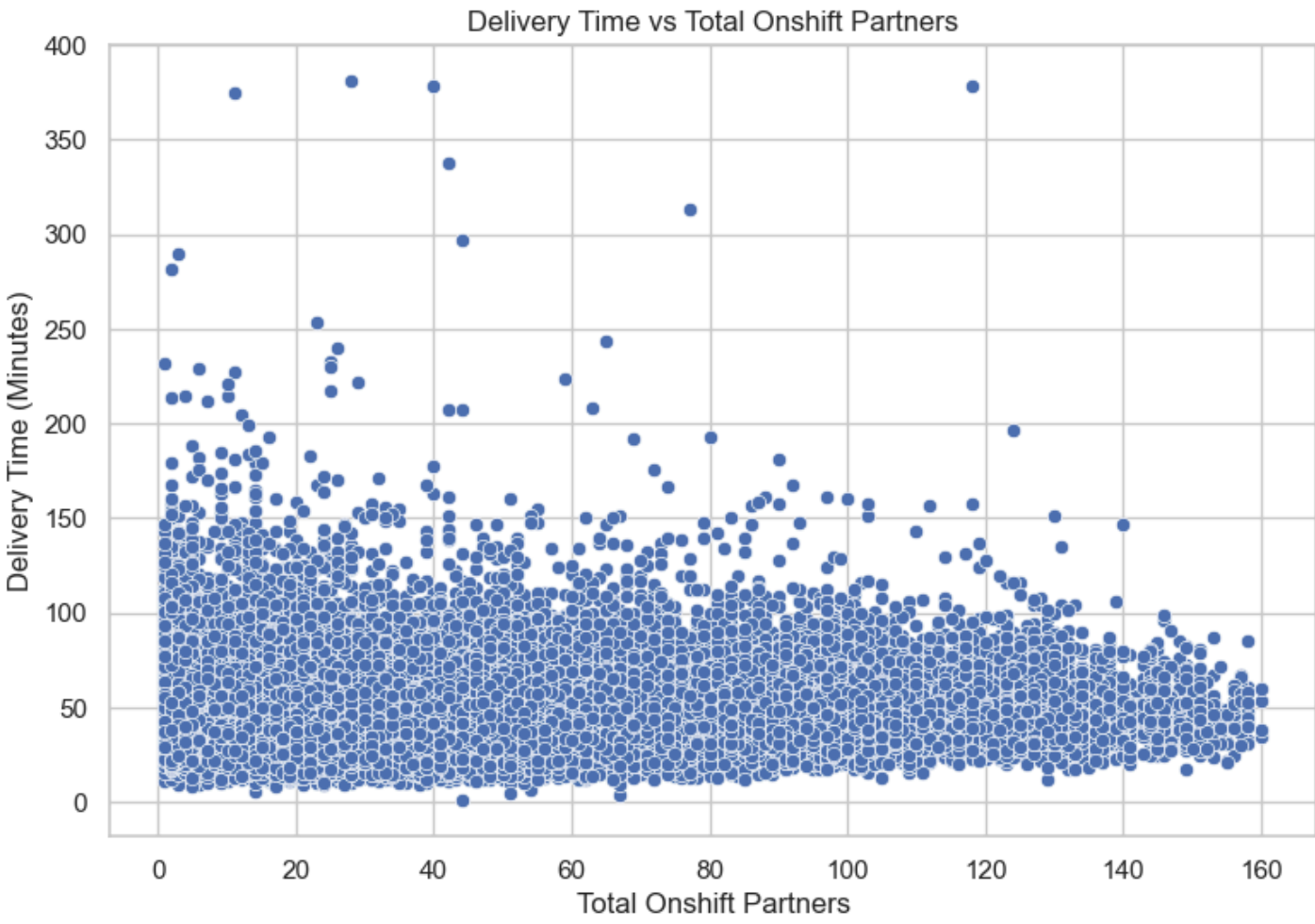
- We can see that the delivery time is not affected by the Store category

```
In [106... fig, ax = plt.subplots(1,2, figsize=(20, 6))

sns.scatterplot(data=t_df, x='total_onshift_partners', y='delivery_time_minutes', ax=ax[0]);
ax[0].set_title("Delivery Time vs Total Onshift Partners");
ax[0].set_xlabel("Total Onshift Partners")
ax[0].set_ylabel("Delivery Time (Minutes)");
```



```
sns.scatterplot(data=t_df, x='total_busy_partners', y='delivery_time_minutes', ax=ax[1]);
ax[1].set_title("Delivery Time vs Total Busy Partners");
ax[1].set_xlabel("Total Busy Partners");
ax[1].set_ylabel("Delivery Time (Minutes)");
```



Observations

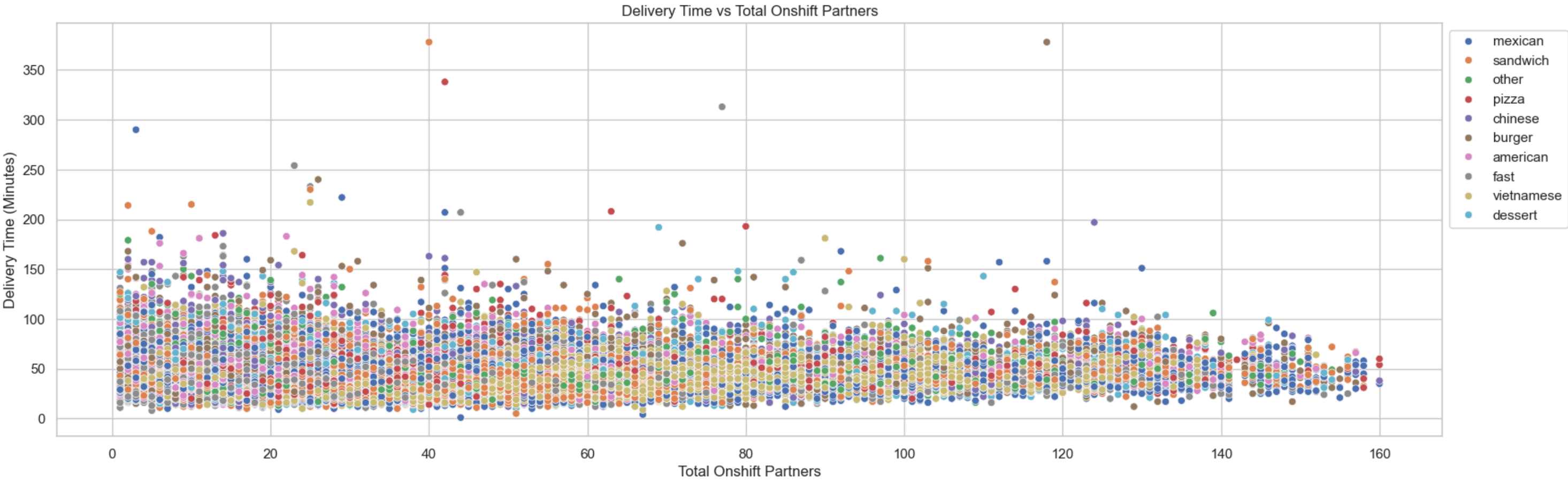
- Delivery time appears to be influenced by the total number of busy partners and the total number of on-shift partners.
- Further analysis is required to understand the relationship and trends.

Multi-Variate Analysis

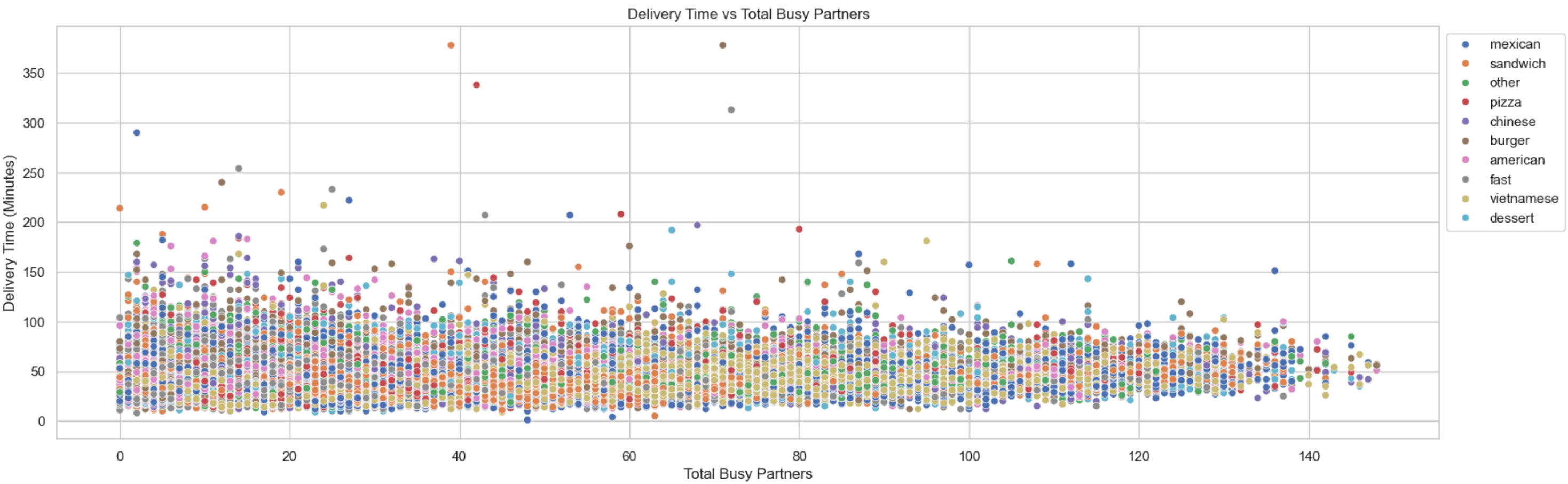
```
In [107...] top_cat=t_df["store_primary_category"].value_counts().head(10).index
top_cat
t_df[t_df["store_primary_category"].isin(top_cat)]
```

...	market_id	created_at	actual_delivery_time			store_id	store_primary_category	order_protocol	total_items	subtotal	num_distinct_items	min_item_price	...	actual_delivery_time
19	1.0	2015-01-30 20:51:23	2015-01-30 21:41:42	ce016f59ecc2366a43e1c96a4774d167		mexican	1	2	950		2	150.0	...	
20	1.0	2015-01-31 18:15:12	2015-01-31 18:44:05	ce016f59ecc2366a43e1c96a4774d167		mexican	1	5	1285		3	150.0	...	
31	3.0	2015-01-30 14:19:30	2015-01-30 14:58:45	50905d7b2216bfeccb5b41016357176b		sandwich	2	4	3490		3	175.0	...	
32	1.0	2015-01-28 15:03:04	2015-01-28 15:34:14	50905d7b2216bfeccb5b41016357176b		other	2	3	1765		3	275.0	...	
33	1.0	2015-02-12 14:57:42	2015-02-12 15:48:29	50905d7b2216bfeccb5b41016357176b		other	2	1	975		1	975.0	...	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
177539	1.0	2015-02-16 18:49:41	2015-02-16 19:54:48	a914ecef9c12ffdb9bede64bb703d877		fast	4	3	1389		3	345.0	...	
177540	1.0	2015-02-12 18:31:59	2015-02-12 19:28:22	a914ecef9c12ffdb9bede64bb703d877		fast	4	6	3010		4	405.0	...	
177541	1.0	2015-01-23 23:16:08	2015-01-24 00:06:16	a914ecef9c12ffdb9bede64bb703d877		fast	4	5	1836		3	300.0	...	
177542	1.0	2015-02-01 12:48:15	2015-02-01 13:53:22	c81e155d85dae5430a8cee6f2242e82c		sandwich	1	1	1175		1	535.0	...	
177543	1.0	2015-02-08 13:54:33	2015-02-08 14:31:41	c81e155d85dae5430a8cee6f2242e82c		sandwich	1	4	2605		4	425.0	...	
52220 rows x 39 columns														

```
In [108...] plt.figure(figsize=(20, 6))
sns.scatterplot(data=t_df[t_df["store_primary_category"].isin(top_cat)], x='total_onshift_partners', y='delivery_time_minutes', hue='store_primary_category');
plt.legend(bbox_to_anchor=(1, 1), loc='upper left')
plt.title("Delivery Time vs Total Onshift Partners");
plt.xlabel("Total Onshift Partners");
plt.ylabel("Delivery Time (Minutes)");
```



```
In [109...] plt.figure(figsize=(20, 6))
sns.scatterplot(data=t_df[t_df["store_primary_category"].isin(top_cat)], x='total_busy_partners', y='delivery_time_minutes', hue='store_primary_category');
plt.legend(bbox_to_anchor=(1, 1), loc='upper left')
plt.title("Delivery Time vs Total Busy Partners");
plt.xlabel("Total Busy Partners");
plt.ylabel("Delivery Time (Minutes)");
```

Observations

- From above plots we cannot see any correlation delivery time, busy delivery partners, and restaurant category

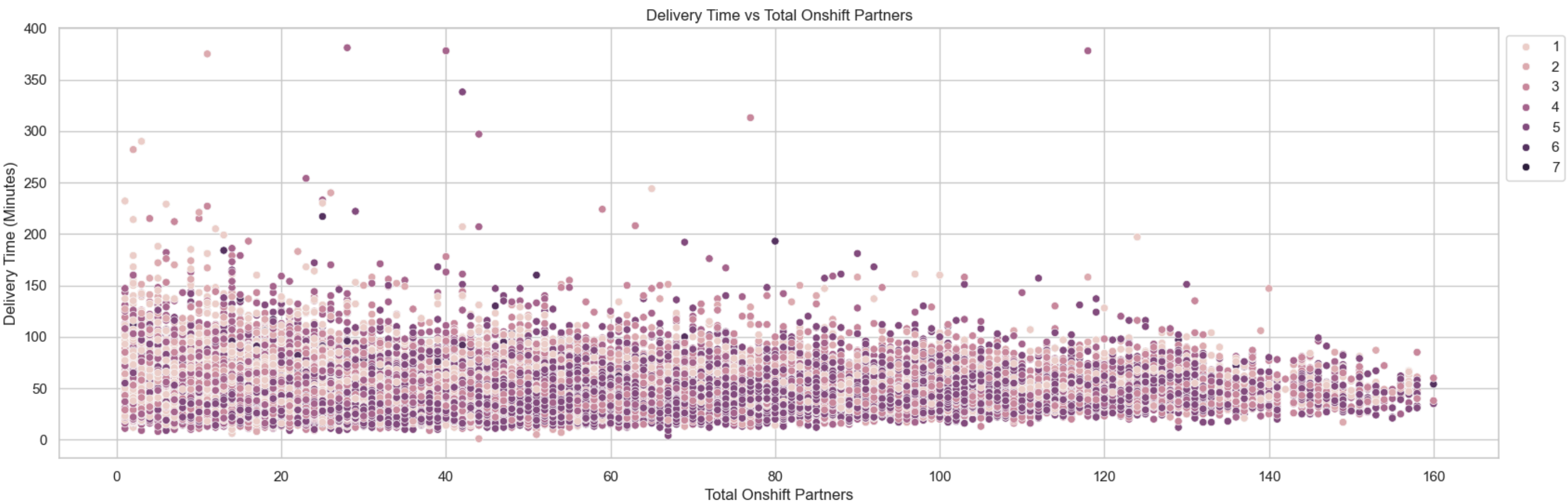
```
In [110... plt.figure(figsize=(20, 6))
sns.scatterplot(data=t_df, x='total_onshift_partners', y='delivery_time_minutes', hue='market_id');
plt.legend(bbox_to_anchor=(1, 1), loc='upper left')
plt.title("Delivery Time vs Total Onshift Partners");
plt.xlabel("Total Onshift Partners")
plt.ylabel("Delivery Time (Minutes)");
```



Observations

- From above plots we cannot see any correlation delivery time, onshift delivery partners, and market id

```
In [111... plt.figure(figsize=(20, 6))
sns.scatterplot(data=t_df, x='total_onshift_partners', y='delivery_time_minutes', hue='order_protocol');
plt.legend(bbox_to_anchor=(1, 1), loc='upper left')
plt.title("Delivery Time vs Total Onshift Partners");
plt.xlabel("Total Onshift Partners")
plt.ylabel("Delivery Time (Minutes)");
```



Observations

- From above plots we cannot see any correlation delivery time, available delivery partners, and order protocol